



Investigating the dynamic relationship between cryptocurrencies and conventional assets: Implications for financial investors[☆]

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ARTICLE INFO

JEL classification:

C22
G01
G14
G11

Keywords:

Bitcoin
Ethereum
Cryptocurrencies
Dynamic relationship
Diversification benefits
Hedging

ABSTRACT

Cryptocurrencies are gradually establishing themselves as a new class of assets with unique features, although there remains skepticism and a lack of understanding of their nature. In this study, we compare the financial properties of these new digital assets and investigate their dynamic relationship with major financial securities and commodities. Furthermore, we evaluate the economic and financial potential benefits of cryptocurrencies for financial investors. Using different time-varying copula approaches and bivariate dynamic conditional correlation GARCH models, we find that the cross-correlation with conventional assets is changing over time but weak, supporting the idea that these cryptocurrencies can be suitable for financial diversification. However, our analysis of portfolios shows that cryptocurrencies are poor hedging instruments in most of the considered cases. Moreover, we find that the relationship between cryptocurrencies and conventional assets is sensitive to external economic and financial shocks.

1. Introduction

Understanding and disentangling the economic and financial properties of cryptocurrencies (CC)¹ remain a challenging and important task for researchers, investors, portfolio managers and regulators (Corbet et al. (2018); Baur et al. (2018)). While our understanding of several of these properties, such as the price determinants of cryptocurrencies, their abilities to act as a safe-heaven and the efficiency of the crypto-markets, has substantially improved in the last few years, (Ciaian et al. (2016); Dyhrberg (2016); Klein et al. (2018)), to date no consensus has been reached on their exact nature and place within the conventional asset classes (Baur et al. (2018)). Even from a regulatory standpoint, this issue is not yet settled. For instance, various U.S. regulatory bodies, with their own areas of oversight, treat cryptocurrencies differently. The U.S. Commodity Futures Trading Commission (CFTC) considers them as commodities, the Internal Revenue Service (IRS) as properties, while the Securities Exchange Commission (SEC) fail to recognize them as securities, despite stressing the

need for better market surveillance and custody for cryptocurrencies. In Canada the authorities at the federal level treat cryptocurrencies as securities, while in Japan and Australia as legal properties. Consequently, determining the true nature of cryptocurrencies and their properties is a decisive topic of research due to its economic and financial implications for policymakers and governments, as well as investors and portfolio managers. Governments are concerned by their negative effects on household wealth and the economic activity in general (European Central Bank (2015)). Governments are also worried about the impacts of this new type of currencies on the overall stability of the monetary and financial system, given that they operate without government or central bank oversight, and their *raison d'être*, is a threat to central banks, as in a world of decentralized currency, the role of a centralized monetary policy is uncertain. For investors and portfolio managers, the question is relatively different since they are particularly interested by the financial benefits that this new type of assets can offer in terms of safe-haven, hedging and diversification.

[☆] We would like to thank the Editor, Professor Mallick, the Associate Editor, the two anonymous referees and the participants to the finance and economics department research seminars at the College of Business and Economics of Qatar University for their helpful comments. Any remaining errors are solely ours.

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¹ CC and digital assets hereinafter are used interchangeably with cryptocurrencies.

The present study aims to explore the dynamic relationship between cryptocurrencies and conventional financial assets, with a particular focus on its implications on the diversification potential of these digital financial assets. Besides the above highlighted issues, there are two other main reasons to conduct this research study. First, this paper is motivated by the convergence of several studies toward the conclusion that cryptocurrencies are relatively disconnected and decoupled from conventional financial assets (See Corbet et al. (2018); Shahzad et al. (2019); Bouri et al. (2017a); Bouri et al. (2017b); Brière et al. (2015)). Multiple factors have been advanced to explain this relative isolation from conventional financial markets. The most important one is that, unlike conventional assets, cryptocurrencies seem to be less dependent on traditional economic and financial variables (Shahzad et al. (2019); Feng et al. (2018); Kristoufek (2015)), and more on their own technological and market development aspects, such as forks, hacking, governments regulations, price speculation, vulnerability to bubbles, and illegal activities financing (Corbet et al. (2018); Feng et al. (2018)). This possible disconnection from conventional assets and exposure to different risk factors would make cryptocurrencies a good diversification tool, which is behind the second motivation. The second motivation is the growing consensus among researchers in the finance literature on the diminishing benefits from diversification² in traditional markets, leading investors to explore new opportunities offered by the prompt growth and increasing development of information and communication technologies (ICTs). The dazzling growth of cryptocurrencies and their identification as a possible new class of assets have gained a significant attention from different profile of investors looking to generate additional portfolio diversification profits. This possibility is of course subject to the relationship of these digital assets with the conventional financial assets, especially given the higher volatility of their prices, and the potential informational inefficiencies of their markets compared to conventional assets (Corbet et al. (2018)).

Despite the growing number of published papers that have explored the different economic and financial aspects of cryptocurrencies, several important questions remain unexplored or pending and need additional investigation. In particular the question of interaction between cryptocurrencies and conventional assets. The answer to this question will pave the way for financial investors, policy makers and regulatory bodies to fully benefit from these new types of digital financial assets. So far, the number of studies investigating the financial dependence structure between cryptocurrencies and conventional assets, as well as their potential diversification benefits are still scarce and limited exclusively to the case of Bitcoin. Furthermore, the few studies that have explored this financial property used standard econometric approaches, not particularly suited for the complex, unstable, and changing nature of cryptocurrencies markets. For instance, applying standard models based on regressions augmented by dummy variables, neither Guesmi et al. (2018), Bouri et al. (2017a), Bouri et al. (2017b), nor Dyhrberg (2016) explore the tail-dependence, nor do they analyze the possible time-varying dependence of Bitcoin and conventional assets. Additionally, none of the only four papers in the literature (Kajtazi and Moro (2018), Klein et al. (2018), Brière et al. (2015), and Guesmi et al. (2018)), effectively assessing the actual diversification benefits and hedging effectiveness of cryptocurrencies goes beyond the case of Bitcoin.

To fill this gap in the literature, this paper adds and contributes to the emerging literature on cryptocurrencies by exploring whether these new digital financial assets can offer new opportunities for portfolio diversification and hedging. To do so, we investigate the dependence

structure between three major conventional financial assets (S&P500, gold and crude oil) and two cryptocurrencies (Bitcoin and Ethereum). In particular, our analysis focuses on the behavior of the time-varying dependence parameter, which is estimated using a battery of time-varying copulas and Dynamic Conditional Correlation GARCH models. Moreover, to better link our results with the political and financial events surrounding the considered set of assets, we test for structural breaks using the Bai and Perron (2003) tests. These results allow us to confirm the time-varying behavior of the dependence parameter and help identify possible financial and international events at the origin of this time-varying dependence pattern. Finally, we quantify the benefits generated by using cryptocurrencies for diversification and/or hedging purposes. Using various time varying copula dependence models (Gaussian, Student-*t*, Gumbel, Rotated-Gumbel, Joe- Clayton and SJC) and three bivariate GARCH Models (BEKK-GARCH, DCC-GARCH, and ADCC-GARCH), we calculate the optimal weights, hedging ratios and effectiveness of several portfolios, including both conventional and/or only cryptocurrencies.

This paper differs from previous studies in many aspects. First, unlike the few papers focusing on the diversification and hedging potential of Bitcoin (Guesmi et al. (2018); Dyhrberg (2016)), our paper extends the digital assets options and includes Ethereum in the analysis. Second, using the most up-to-date dataset, the period covered in this paper includes the bubble boom and burst period, from October 2017 to April 2018. Third, to effectively assess the digital assets diversification potential and hedging capabilities and effectiveness, we construct various portfolios, pairing a digital with a traditional asset, as well as a pure digital assets portfolio. In this regard, except for Guesmi et al. (2018), we depart from most of the previous studies that restrict their analysis to the possibility of using Bitcoin as a diversifying and/or hedging instrument, without investigating the optimal weight allocation or the actual hedging effectiveness of the digital asset (Kajtazi and Moro (2018), Klein et al. (2018), Brière et al. (2015), use several measures portfolio performances. See also Corbet et al. (2018), Bouri et al. (2017a), and Dyhrberg (2016)).

The choice of Bitcoin (BTC) and Ethereum (ETH) as digital financial assets is motivated by several considerations.³ First, among the hundreds currently traded cryptocurrencies, BTC and ETH have attracted a considerable devotion from financial market participants, investors and portfolio managers given their innovative aspect, immense flexibility, complete decentralization and their rapid prices and market capitalization growth. As of October 2018, in a crypto-market formed by more than 2112 CC (coins and tokens) and 15,124 exchanges, the market share of these two CC exceeds 50% of the total CC market (see Fig. 1). Moreover, the returns of these two blockchain-based cryptocurrencies have seen large increases which, in several cases, have exceeded the level of 100% in the day-to-day returns. The Bitcoin price increased by approximately 7,309,666% between July 2010 and October 2018, from 0.09 USD to 6578 USD, and the Ethereum price has increased by more than 17,115% since its first trading day on August 7th, 2015, from 1.33 USD to 228.96 USD on October 1st, 2018.⁴ Among the current top 5 cryptocurrencies, Bitcoin and Ethereum are also the most active; in

² There is a strong consensus among researchers, financial investors and portfolio managers that the benefits of diversification are actually very limited due mainly to the increasing globalization and growing integration of financial markets. See Berger et al. (2011); Arouri and Rault (2012); Balli et al. (2013); Narayan and Islam (2014); Charfeddine (2016); Balcilar et al. (2015); Mimouni et al. (2016).

³ Our choice not to include Ripple, the third top Cryptocurrency, is motivated by the fact that it is conceptually different from BTC and ETH. Unlike the mined BTC and ETH, Ripple, whose supply is controlled by the Ripple company, aims to be a link between fiat currencies during a cross-border transaction and is not competing with fiat currencies. Moreover, the non-inclusion of the fourth top Cryptocurrency, namely, Bitcoin Cash, is due to data availability, since this Cryptocurrency was introduced after a hard fork, an upgrade in the Blockchain that is not backward compatible, on August 17th, 2017.

⁴ These extreme changes are at the center of a literature exploring the price formation of Bitcoin. Besides speculation (Bouoiyour and Selmi (2015)), this volatility is partly due to the fact that cryptocurrencies are detached from the real economy, implying that there are no macroeconomic fundamentals that would determine their price formation (See Ciaian et al. (2016)).

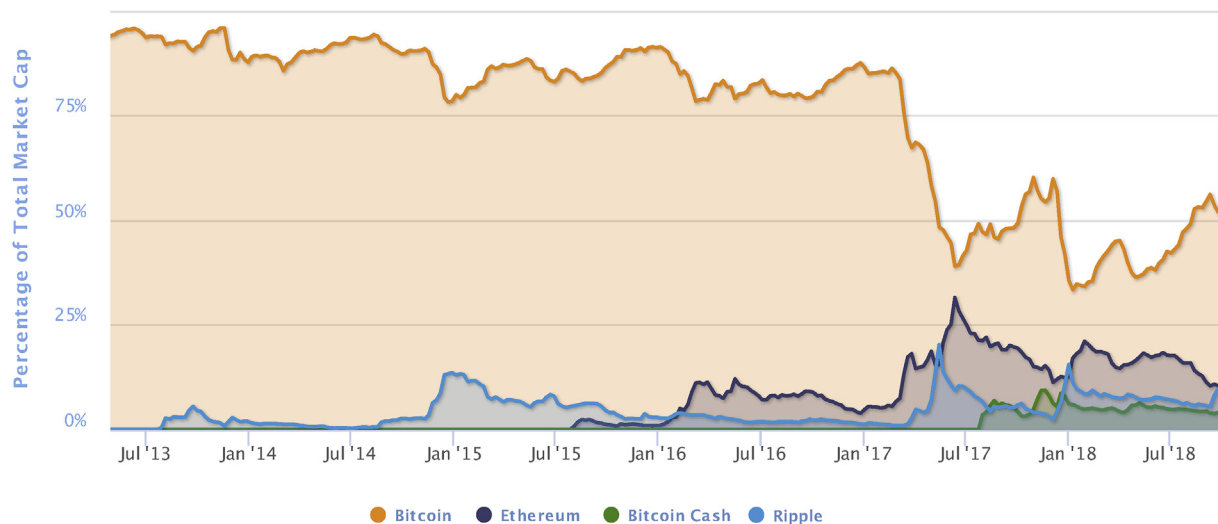


Fig. 1. Market share of the top 4 CC in term of market capitalization.

Source: <https://coinmarketcap.com/charts/>

2018, an average daily transaction of 200,000 for Bitcoin and 500,000 for Ethereum have been recorded, compared to less than 50,000 daily transactions for Ripple, Bitcoin Cash and Litecoin.

The results of this empirical study have several important implications for investors and policymakers in terms of diversification and hedging strategies, as well as the relationship of cryptocurrencies with conventional assets. First, the results show that while the dependence between digital and conventional assets is weak, it is important between the two considered cryptocurrencies. Second, the results of applying the dynamic copula approach show evidence of time varying dependence between the seven examined pairs, a result which is confirmed by the Bai and Perron (2003) tests for structural breaks. In particular, the results show evidence for an increasing dependence between the two financial digital assets over time. Third, the results show that the weights of cryptocurrencies necessary for optimal portfolio diversification are small in general and do not exceed the level of 18.41%, which corresponds to the portfolio composed of BTC-Crude oil. For the pair BTC-ETH, the weight of BTC does not exceed 25.13%. Fourth, the results indicate that the cryptocurrencies are poor hedging instruments. Finally, we did not find any short or long run dependence in the marginal distribution (series of returns), indicating that all the conventional and digital assets are efficient.⁵

The remainder of this paper is organized as follows. Section 2 presents a summary of the related literature, Section 3 briefly presents the emergence and the development of cryptocurrencies. Section 4 presents the empirical methodology, with a focus on static and time varying copula models. Section 5 presents and discusses the empirical findings of this study in terms of time-varying dependence behavior. Section 6 presents the results regarding the diversification, hedging ratios and hedging effectiveness of CC. Finally, section 7 concludes and discusses some of the implications of our results.

2. Literature review

The cross-asset correlation (dependence) property between the returns of different financial assets is one of the most examined property in the finance literature (Cont (2001), Engle (2002),

Cherubini et al. (2004), and Adams et al. (2017)). Indeed, the level of dependence between the returns of different financial assets can inform investors about the different financial capabilities of the underlying assets such as their safe-haven, hedging, and diversification properties.

The interaction between cryptocurrencies and the conventional assets is at the center of a growing literature. Most of the empirical studies investigating these dependence properties focus on the safe-haven and hedging abilities of cryptocurrencies (Shahzad et al. (2019); Feng et al. (2018); Klein et al. (2018); Bouri et al. (2017a); Bouri et al. (2017b); Dyhrberg (2016)). In addition, the number of studies effectively examining the actual benefits of cryptocurrencies in terms of diversification and hedging (Guesmi et al. (2018); Kajtazi and Moro (2018)) is still scarce and limited. Furthermore, the literature explores mainly the case of Bitcoin, with only few recent studies including other altcoins to their analysis (Corbet et al. (2018); Baumöhl (2018); Feng et al. (2018)).

The studies investigating the relationship between stocks, bonds, and cryptocurrencies have focused solely on Bitcoin. Overall, they found that Bitcoin is uncorrelated (Dyhrberg (2016)) or weakly correlated to stocks (Bouri et al. (2017a); Guesmi et al. (2018), Baur et al. (2018), Fang et al. (2019), Briere et al. (2015)). Regarding the nature of this relationship, Klein et al. (2018) find that it is characterized by quickly alternating phases of positive and negative correlations. Shahzad et al. (2019) find that the relationship is characterized by time-variability and non-linearity with a positive correlation in market turmoil. Furthermore, Corbet et al. (2018) suggest that cryptocurrencies are not connected to conventional markets and that are not affected by the fundamental macroeconomic indicators. Some of the reasons behind the weak relationship between Bitcoin and stock markets, include the fact that they have different pool of investors, where stocks markets involve more institutional investors as compared to Bitcoin with younger investors with different strategies (Shahzad et al. (2019); Ji et al. (2018)). It was also advanced that Bitcoin does not share common price determinants with conventional financial assets ((Bouoiyour and Selmi (2015); Kristoufek (2015)). For instance, the price of Bitcoin is less dependant on economic and financial variables, but is determined by a set of unique factors such as investors sentiment and attractiveness (Kristoufek (2015)), energy prices (Bouri et al. (2017b)), user anonymity, technology, and criminal activities (Ciaian et al. (2016), Kristoufek (2013), and Kristoufek (2015)).

Overall, there is some consensus in the literature that Bitcoin is a good diversifier for stocks and bonds. However, for the remaining

⁵ The modelling of the marginal distribution is a necessary step for both the dynamic copula approach and multivariate GARCH models.

altcoins, and to the best of our knowledge only two studies, Corbet et al. (2018) and Fang et al. (2019), have explored the relationship between stocks and bonds, and cryptocurrencies besides Bitcoin. Fang et al. (2019) explored the tail dependence between seven selected cryptocurrencies and four stocks indices. They found that cryptocurrencies are both left tail and cross tail independent from the four selected stock indices. Focusing only on three cryptocurrencies (Bitcoin, Ripple and Litecoin), Corbet et al. (2018) use connectedness measures in the time and frequency domains and find that cryptocurrencies are isolated from the S&P 500 index and Markit ITTR110 index for bonds. This isolation, they note, is a sign that cryptocurrencies can offer new opportunities for financial diversification.

Bitcoin has also been vastly compared to commodities, particularly to Gold, as intended by its inventor. Besides the conceptual similarities of “mining”, Bitcoin and Gold share many characteristics. Even if they do not make claims on future cash flows, their value is linked to their supply and demand. Like Gold, Bitcoin is “mined”, and its supply is limited by its protocol to 21 million coins (Böhme et al. (2015)), and no central authority can control or adjust their actual supply (Baur et al. (2018)). Bitcoin is found to possess some of the same hedging capabilities as gold (Dyhrberg (2016)). However, unlike gold, Bitcoin is a poor hedge and is suitable for diversification purposes only (Bouri et al. (2017a)), as its hedging and safe-heaven properties vary between horizons. Despite the various studies comparing Bitcoin to Gold in respect to other financial assets, the nature of their relationship seems to be ignored by most of the literature. Among the few papers analyzing the relationship between Bitcoin and Gold, Brière et al. (2015) found a significant relationship between Gold and Bitcoin among other assets. The positive correlation is also confirmed by Klein et al. (2018). Besides Gold, Bitcoin has also been compared to silver by Klein et al. (2018), the authors found that Bitcoin, like the two metals (Gold and Silver), exhibit an “Inverse Leverage Effect”, that is, the implied volatility rise when their prices increase. This observation contradicts the findings of Dyhrberg (2016), who does not observe an “Inverse Leverage Effect”, but reports an insignificant leverage effect for Bitcoin, that is the volatility declines (rises) when returns increase (decrease).

Another commodity which is related to Bitcoin and cryptocurrencies is energy. Indeed, the cryptocurrencies relying on mining are big energy consumers, as energy is an essential input for those mined cryptocurrencies. The link between cryptocurrencies and energy has been established for the case of Bitcoin (see Bouri et al. (2017b)), suggesting that Bitcoin value reflects the cost of production, dominated by energy consumption, and that the lower-bound of Bitcoin’s fundamental value is determined by the cost of the energy involved in its mining (Garcia and Schweitzer (2015)). However, Ciaian et al. (2016) provide evidence that oil price has a significant impact on the price of in the short run only, while Baur et al. (2018) found that Bitcoin returns are uncorrelated with oil or gold commodities both in normal times and in periods of financial turmoil. In the meantime Bouri et al. (2017a), show that Bitcoin is negatively correlated with the commodity index but the link vanishes when they consider weekly data. The time-varying relationship between commodities and Bitcoin is confirmed for energy by Bouri et al. (2017b) who report that Bitcoin and global energy commodity become positively but weakly correlated only after 2013, within their sample covering the period 2010–2015. (See also Selmi et al. (2018)).

The rise of interest in alternative digital currencies has seen the emergence of few papers exploring the relationship between different altcoins (Baumöhl (2018), Corbet et al. (2018), and Platanakis et al. (2018)). Phillip et al. (2018) try to highlight the different features defining cryptocurrencies, and show that the 224 cryptocurrencies considered in their study exhibit long memory, leverage, stochastic volatility and heavy tailedness. Brauneis and Mestel (2018) test the efficiency of several cryptocurrencies and additionally link their efficiency to measures of liquidity. They conclude that most CC become less inefficient as liquidity increases. Corbet et al. (2018) explore the dynamic relation-

ships between three cryptocurrencies, Bitcoin, Litecoin and Ripple and find that cryptocurrencies are highly connected to each other. Considering the case of four cryptocurrencies, Platanakis et al. (2018) show a relative low correlation between Ripple and Dash, but find a relatively high correlation between Bitcoin, Litecoin, Ripple and Dash. This result contradicts Baumöhl (2018) who show that the connection between cryptocurrencies is on average not as strong as is widely believed. However, we should note that these studies consider different cryptocurrency pairs. For instance, Platanakis et al. (2018) does not consider the second largest cryptocurrency, Ethereum. Taking into account Ethereum, Baumöhl (2018), contrary to Corbet et al. (2018), finds that ETH is significantly positively associated with BTC. Besides the inconclusive results of these studies, none of them effectively assesses the practical implications of these links between the respective considered CC.

From this overview of the related literature, three important gaps emerge. First, at the expense of the altcoins, most of the studies use Bitcoin to explore the interaction between cryptocurrencies and conventional assets. Second, the number of studies assessing the actual diversification benefits and hedging effectiveness of cryptocurrencies are scarce and limited. Additionally, this scant literature reaches indecisive results. Guesmi et al. (2018), confirm that adding Bitcoin to a portfolio reduces considerably its risk. Klein et al. (2018), however find that Bitcoin does not provide a hedge for equity investments, and like Kajtazi and Moro (2018), Brière et al. (2015), note that including Bitcoin to a portfolio improves the portfolio returns, but tends to increase its volatility.

3. The CC market

In October 2008, the enigmatic Satoshi Nakamoto published a paper on the Cryptography Mailing list at metzdowd.com, *Bitcoin: A Peer-to-Peer Electronic Cash System*. In that paper, the author(s) laid down the foundation of a cryptocurrency, Bitcoin, and the technology underlying it, the Blockchain, which is believed to bring a novel solution to a problem that has been an issue for thousands of years: trust in value exchange. The originality of Bitcoin and the Blockchain is transferring the process of trust creation to a collective agreement body of independent computers instead of a single central third party. The cryptocurrencies appeared as an alternative and a response to the continuous manipulation of fiat money by central banks. As stated by Nakamoto, “The root problem with conventional currency is all the trust that is required to make it work. The central bank must be trusted not to debase the currency, but the history of fiat currencies is full of breaches of that trust ... With e-currency based on cryptographic proof, without the need to trust a third-party middleman, money can be secure and transactions complete.” (Dowd and Hutchinson (2015)).

Technically, Bitcoins represent transactions registered on a public open ledger, with every new transaction to be approved by the participants (known as miners) and added to a block which will be then linked to a chain comprising all the previous blocks (for a detailed explanation of Bitcoin’s architecture and its economic and governance consequences, see, for instance, Böhme et al. (2015)). Driven by the success of Bitcoin, a wide range of other CC, known as altcoins, grew to represent as much as of 60% of cryptocurrencies’ market capitalization. The most well-known and second most important in terms of market capitalization after Bitcoin is Ethereum. Ethereum is a more sophisticated network, as it can also store applications - computer codes - which can then be powered by the computing power going into the network formed by all the computers connected to it. This computing power represents the currency, Ether, which allows its holders to use the resources provided by the network to run their applications. One of these applications are automated agreements, known as Smart Contracts, which are executed when the conditions embedded in them are recognized as being met by the network.

Table 1
Keys figures about the top 4 CC as of October 1st, 2018

Name	Market Cap. (MC)	Share in MC	Circulating Supply	Maximum Supply
Bitcoin	\$114,509,724,600	51.59%	17,296,357 BTC	21,000,000 BTC
Ethereum	\$23,855,852,479	10.61%	102,273,607 ETH	Unlimited
Ripple	\$23,265,129,410	10.25%	39,870,907,279 XRP	100 billion
Bitcoin Cash	\$9,238,550,523	4.10%	17,376,231 BCH	21 Million

Notes. This table presents some keys figures about the top 4 CC in term of market capitalization. It reports information about the market capitalization, share in in the CCs market, circulating supply and maximum supply as of October 1st, 2018.

Source: coinmarketcap.com/

Ten years after Nakamoto's paper, Bitcoin and other cryptocurrencies evolved to become a global phenomenon that has attracted substantial attention from the media and the general public. The staggering price movements and rapid increase of cryptocurrencies market capitalization are among the most important factors explaining this growing popularity (Garcia and Schweitzer (2015)). For instance, Bitcoin was ranked number two in the 2017 Google Trend list for the global news category, just after Hurricane Irma. On Wikipedia, as of November 2018, the Bitcoin page has been viewed 27,125,944 times, with an average of 22,108 daily views, since July 2015. Moreover, between May 1st, 2013 and October 1st, 2018, the global market capitalization of CC increased from 1.344 billion US dollars to 220.723 billion US dollars, which corresponds to an average annual increase of approximately 156%.⁶ The price of Bitcoin increased by more than 21 Million % between 18/07/2010 and 16/12/2017 (from 0.09 USD per Bitcoin to 19,343 USD), to decrease by 65% between 16/12/2017 and 01/10/2018. Similar to Bitcoin, the price of Ethereum has increased by more than 103,826% between 01/09/2015 and 14/01/2018 and decreased by more than 80% between 14/01/2018 and 01/10/2018⁷ (see Table 1 and Fig. 1 for the top 4 Cryptocurrencies).

The CC market is, however, still risky, as investors do not have any legal recourse in case of fraud or failure. The powerful investment opportunities offered by CC have been tainted by the different scam and fraud scandals. If CC are still allowed and unregulated in many places, they have been banned in countries such as China and South Korea. China wants to completely eradicate cryptocurrencies by blocking access to any cryptocurrency, although Chinese investors have been among the biggest actors in the cryptocurrencies markets. The question of regulating cryptocurrencies is still a hot topic. While there seems to be a consensus among regulators for the need to regulate CC markets, the form of such regulations is still to be defined. This is because cryptocurrencies are not yet fully understood and well defined. Treating them as currencies or assets will lead to different regulations.

Numerous events, such as the bankruptcy of the biggest CC exchange, Mt. Gox, in 2014, or the lack of consensus among the developers community that leads to different branches of Bitcoin, for instance, are factors jeopardizing the stability of the CC markets and add a layer to the uncertainty surrounding them. The International Monetary Fund (2018) mentions that: "It is impossible to know the extent to which crypto assets may transform the financial infrastructure and whether most new crypto assets are likely to disappear as in past episodes of technological innovation (as many tech companies did during the boom of the late 1990s, for example). Before they can transform financial activity in a meaningful and lasting manner, crypto assets will first need to earn the confidence and support of consumers and financial authorities."

⁶ Note that the CC market capitalization has reached its highest level of 828.537 billion USD dollars in January 07th, 2018. Data source: <https://coinmarketcap.com/>.

⁷ The dataset used in this study covers all the periods where the two cryptocurrencies were traded.

4. Empirical methodology

In this section, we introduce the materials and methods used to investigate the time varying dependence structure between digital and conventional financial assets. We start by presenting in the next subsection, 4.1, the ARFIMA – FIAPARCH models used to model the marginal distribution of the different returns series. Then, in subsection 4.2, we present the time-varying copula techniques used to model the dependence between the different returns series. Finally, in subsection 4.3, we present the structural changes tests employed to test for regimes changes in the estimated time-varying dependence parameter. The robustness of the empirical findings is examined by employing three bivariate GARCH models.⁸

4.1. Marginal distribution

A common step for both static and time varying copula approaches is to model the marginal distribution of each return series before a copula estimation. In fact, the dependence structure between the two series of returns will be correctly captured if the marginal distributions are well-specified. An important condition when modelling the univariate distribution is the necessity to obtain the independently and identically distributed (i.i.d.) residuals that are required for the copula model estimation (see for instance, Grégoire et al. (2008)).

In this study, we propose to model the marginal distribution of the considered series of returns by using the ARFIMA(p, d_m, q) – FIAPARCH(k, d_v, l) model which nest a large number of specifications.⁹ The use of the ARFIMA(p, d_m, q) – FIAPARCH(k, d_v, l) model is motivated by the large varieties of stylized facts that this model can account for, including volatility clustering, asymmetries of shocks and long memory behaviors. Moreover, several studies have shown the superiority of this class of models in modelling the returns series for the case of conventional financial assets (Ftiti and Chaouachi (2018); Jawadi et al. (2017); Charfeddine (2016); Charfeddine and Benlagha (2016); Abuzayed et al. (2018)). For the digital assets, while the empirical literature is very recent, several studies have used the GARCH class of models to account for the presence of volatility clustering, fat tails, asymmetries of shocks as well as the long memory behavior observed in both the return and volatility series¹⁰.

⁸ The bivariate GARCH models used in this study include the diagonal BEKK-GARCH of Engle and Kroner (1995), DCC-GARCH of Engle (2002) and ADCC-GARCH of Cappiello et al. (2006). To save space, these three bivariate GARCH models are not presented here.

⁹ The nine marginal distributions are the ARFIMA(p, d_m, q) – GARCH(k, l), the ARFIMA(p, d_m, q) – GARCH(k, l), the ARFIMA(p, d_m, q) – APARCH(k, l), the ARFIMA(p, d_m, q) – FIGARCH(k, d_v, l), and the ARFIMA(p, d_m, q) – FIAPARCH(k, d_v, l) models.

¹⁰ In a recent study, Charles and Darné (2019) question the ability of the GARCH-class of models to capture all the financial properties observed in Bitcoin returns.

4.1.1. The mean equation: an ARFIMA(p, d_m, q) model

The ARFIMA(p, d_m, q) process is used to model the mean equation of the return series. One of the advantages of using this process is its ability to account for both short and long-run dependence between observations. In this model, the long-run dependence is captured by the fractional parameter of long memory, d_m , and the short-run dependence by the autoregressive components AR(p). The optimal lag length of the orders, p , of the autoregressive component, AR(p), and, q , of the moving average component MA(q) are determined by using the AIC criteria.

The general form of the ARFIMA(p, d_m, q) process is given by the following:

$$\Phi(L) (1 - L)^{d_m} (r_t - \mu) = \Theta(L) \varepsilon_t \quad (1)$$

where L is the lag operator, $\Phi(L) = 1 - \sum_{i=1}^p \varphi_i L^i$ and $\Theta(L) = 1 + \sum_{j=1}^q \theta_j L^j$. The φ_i and θ_j for $i = 1, \dots, p$ and $j = 1, \dots, q$ are the parameters to estimate, and $d_m = d_{ARFIMA}$ is the long-memory parameter of the mean equation.

4.1.2. The variance equation: A FIAPARCH(k, d_v, l) model

To account for long range dependence and asymmetry in the volatility series, we propose using the FIAPARCH(k, d_v, l) model, as developed by Tse (1998). Formally, the FIAPARCH(k, d_v, l) can be expressed as follows:

$$\varepsilon_t = h_t \vartheta_t \quad (2)$$

$$h_t^\delta = \omega[1 - \beta(L)^{-1}] + \{1 - [1 - \beta(L)^{-1}]\phi(L)(L - 1)^{d_v}\}(|\varepsilon_t| - \gamma\varepsilon_t)^\delta \quad (3)$$

where the error term ϑ_t follows either a Gaussian or the Student's t -distribution where τ degrees of freedom and h_t denotes the time-dependent conditional variance.

Under the FIAPARCH model, the volatility is characterized by a long-range dependence behavior if the parameter d vary between 0 and 1. δ is the power term in the volatility structure that takes positives values and γ is the parameter of asymmetry. The term $(|\varepsilon_t| - \gamma\varepsilon_t)^\delta$ reflects the leverage effect.

4.2. Static to time-varying copulas

In a portfolio formed by two assets (one digital and one conventional), all the dependence is enclosed in the joint density. If the entire joint distribution of the returns is known, then one could capture all the relevant dependencies and, consequently, explore the possible existence of diversification opportunities. However, in practice, the determination of the structure of dependence is usually a challenging task, especially for large portfolios. Indeed, there might not be a unique parametric joint density characterizing the relationship across all variables. Furthermore, the estimation of the joint density can lead to misspecification errors if a parametric method is used to estimate the joint density (Chollete et al. (2011)). To overcome these estimation issues, copula functions, initially introduced by Sklar (1959) and widely used in Finance and Economics (see, e.g., Cherubini et al. (2004); Benlagha (2014); McNeil et al. (2015)), provide more flexible measures of the dependence structure between asset returns and may help shed light on diversification opportunities.

Actually, the dependence among financial assets does not remain constant over time. The time-varying nature of the comovements and dependence is widely documented (see Patton (2006) and Longin and Solnik (2001)). The common deduction from these studies is that the dependence and comovements in stock returns evolve over time due to high economic and financial shocks, financial regulations, government announcements, financial instability and crises. Hence, the level of dependence usually changes before, during, and after bad or good news (see Grundke and Polle (2012) and Adams et al. (2017)). For example, a strong correlation before a financial crisis or economic turmoil can be followed by a significant decoupling during the crisis. To

account for the possible existence of this behavior, Patton (2006) proposed a new methodological approach that extends Sklar's theorem and introduces the conditional copula function to model time-varying conditional dependence.

Let $r_{d,t}$ and $r_{c,t}$ denote digital asset returns (Bitcoins or Ethereum) and conventional financial asset returns at time t , respectively. Their respective marginal conditional cumulative distribution function are $u_{d,t} = G_{d,t}(r_{d,t} | \Psi_{t-1})$ and $u_{c,t} = G_{c,t}(r_{c,t} | \Psi_{t-1})$ where Ψ_{t-1} reflects the past information. Formally, the conditional copula function $C_t(u_{d,t}, u_{c,t} | \Psi_{t-1})$ can be written using the two time-varying cumulative distribution function. The bivariate conditional cumulative distribution functions of random variables $r_{d,t}$ and $r_{c,t}$, as extension of Sklar's theorem, can be presented as follows:

$$F(r_{d,t}, r_{c,t} | \Psi_{t-1}) = C(r_{d,t}, r_{c,t}) \quad (4)$$

The conditional joint density can be written as:

$$\begin{aligned} f(r_{d,t}, r_{c,t} | \Psi_{t-1}) &= \frac{\partial F(r_{d,t}, r_{c,t} | \Psi_{t-1})}{\partial r_{d,t} \partial r_{c,t}} \\ &= c_t(r_{d,t}, r_{c,t} | \Psi_{t-1}) \times g_{d,t}(r_{d,t} | \Psi_{t-1}) \\ &\quad \times g_{c,t}(r_{c,t} | \Psi_{t-1}) \end{aligned} \quad (5)$$

where $c_t(u_t, v_t | \Psi_{t-1}) = \partial^2 C_t(u_t, v_t | \Psi_{t-1}) / \partial u_t \partial v_t$ is the conditional copula density function and $g_t(\cdot)$ is the density function corresponding to $G_t(\cdot)$.

The likelihood function issued from Eq. (5) can be written as follows:

$$L_{d,c}(\Theta) = L_d(\Theta_d) + L_c(\Theta_c) + L_{cop}(\Theta_{cop}) \quad (6)$$

where Θ_d and Θ_c are the parameter vectors of marginal distributions of the digital and conventional asset returns, respectively, and Θ_{cop} is the vector of parameters in the copula function. Following Patton (2006), the dynamics of the time-varying dependence between digital asset returns (Bitcoins or Ethereum) and conventional asset returns in the conditional Gaussian and Student- t copulas is specified as follows:

$$\rho_t = \tilde{\Lambda} \left(\psi_0 + \psi_1 \cdot \rho_{t-1} + \psi_2 \frac{1}{q} \sum_{j=1}^q \Phi^{-1}(u_{t-j}) \Phi^{-1}(v_{t-j}) \right) \quad (7)$$

where $\tilde{\Lambda} = (1 - e^{-x})(1 - e^{-x})^{-1}$ corresponds to the logistic transformation. The purpose of employing the logistic transformation is to ensure that the value ρ_t is comprised in $(-1, 1)$. This equation assumes that the copula dependence parameter follows an ARMA(1, q) type process in which ψ_0 is a constant, and $\psi_1 \cdot \rho_{t-1}$ corresponds to the autoregressive term capturing the persistence effect, and the last component reflects the variability of the dependence, described by the last q observations' average product of the transformed variables. The same specification is employed for the Student- t copula by replacing $(\Phi^{-1}(\cdot))$ by $(t_v^{-1}(\cdot))$.

4.3. Structural change tests

To further analyze the structure of the dependence parameter, in this subsection, we apply the Bai and Perron (2003) tests of structural changes on the different time series of the estimated parameter of dependence. By employing these tests, we attempt to explore whether the mean of the parameter of dependence is significantly different from zero and whether the dates of shifts coincide with specific financial and/or international events related to the digital and/or conventional financial markets.

The model on which the Bai and Perron (2003) tests for structural breaks are applied is given by the following equation:

$$\widehat{dep}_t = c_j + u_t \quad (8)$$

where m is the number of breaks, $t = T_{j-1} + 1, T_{j-1} + 2, \dots, T_j$, $j = 1, 2, 3, \dots, m + 1$, $T_0 = 0$, and $T_{m+1} = T$. c_j is the mean of the estimated dependency parameter for each regime. As recommended by

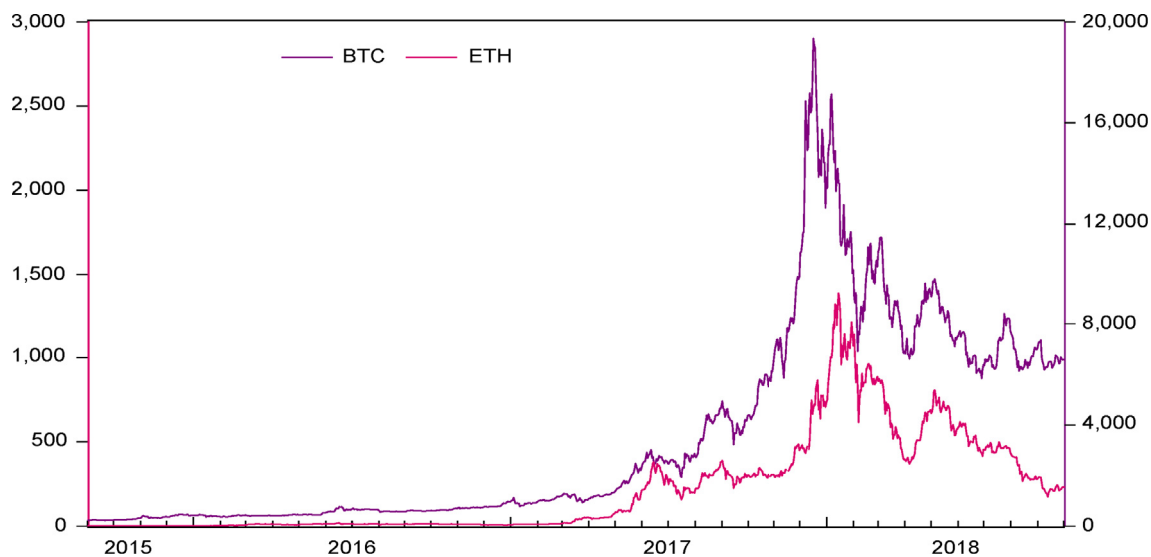


Fig. 2. Bitcoin and Ethereum price's evolution.

Bai and Perron (2003), we use the sequential test of the $L + 1$ vs L breaks testing procedure. However, before applying this test, one should define a set of parameters such as the trimming and the maximum number of breaks. In this paper, as suggested in many previous studies, we set the trimming parameter to 15% and the maximum number of breaks, m , to five breaks. Moreover, we allow for the errors terms, u_t , to have different distributions across the regimes.

5. Results and discussion

5.1. Data description and preliminary analysis

Due to the large number of active Bitcoin and Ethereum exchanges and the multiplication of websites providing aggregated information on cryptocurrencies, each using its own methodology,¹¹ and to keep our data-set consistent, we collected our data from <https://coindesk.org>. The covered periods span from July 18th, 2010 to October 1st, 2018 for Bitcoin and from September 1st, 2015 to October 1st, 2018 for Ethereum.¹² For the three conventional financial assets, we use the two most traded energy and metal commodities namely, Gold and Crude Oil, as well as the most used stock market index in the financial modelling literature, i.e., S&P 500. The data regarding these three variables, was extracted from the Federal Reserve Bank of St. Louis database, <https://fred.stlouisfed.org/>. All the returns series are calculated in a continuous compound basis, $r_{i,t} = 100 * \log(P_{i,t}/P_{i,t-1})$, where $P_{i,t}$ is the closing price of asset i (BTC, ETH, S&P 500, Gold, and Crude Oil) at time t . As volatility proxy, we use the squared returns series.

We begin our preliminary analysis of the cryptocurrencies and conventional financial assets by graphically examining their price evolution as reported in Figs. 2 and 3. We observe that, in contrast to the price of the three financial assets, the evolution of the cryptocurrencies' price exhibits the presence of several periods of sharp increases

and decreases. In particular, we observe that the dates of these sharp increases and decreases coincide with some financial and international events related to the considered assets in this study (see Figs. 14 and 15). We continue the analysis by comparing the returns and volatility of the digital and conventional assets based on their box-plots reported in Figs. 4 and 5, respectively, and the trajectories of the returns series (See Figs. A1 and A2 in the supplementary document). Figs. 4 and 5 clearly show that the cryptocurrencies are highly volatile and characterized by high levels of returns compared to conventional financial assets. In contrast, the series of returns trajectories show that both conventional and digital financial share some well-known financial stylized facts, such as the presence of volatility clustering and asymmetric drawdowns and drawups movements of returns. This graphical preliminary analysis indicates that cryptocurrencies and the considered conventional assets may fall into two different classes of financial assets. Several reasons can explain this high differences in terms of price, returns, and volatility (Shahzad et al. (2019), and Garcia and Schweitzer (2015)). One of these reasons is that the cryptocurrencies market is not mature, compared to the markets of conventional assets. The cryptocurrency market is also more prone to investors speculation and speculative bubbles, which create sharp jumps in cryptocurrencies prices, generating high levels of returns and variability. Moreover, cryptocurrencies are subject to different governments regulations, bans, high risks of fraud and forks among many other factors likely to impact their prices (Shahzad et al. (2019), Ciaian et al. (2016), Garcia and Schweitzer (2015), and Kristoufek (2015)). In contrast, conventional assets markets are well established and regulated, less prone to manipulations and assets prices are more linked to the fundamentals of the economy and more likely to reflect the assets' intrinsic value.

Besides the graphical analysis, we report in Table 2 a summary of descriptive statistics for all the considered assets' series of returns; for both samples I and II.¹³ Table 2 confirms most of the above conclusions based on the graphical analysis. The results show that while the average daily returns is quiet small and close to zero for the three conventional financial assets (0.012, 0.000, and 0.038 for S&P500, Gold and Crude Oil, respectively), it is positive and extremely high for the two digital financial assets (0.130 for Bitcoin and 0.20 for Ethereum).

¹¹ Most of the prices provided by these websites represent an average of cryptocurrencies' prices in the top active exchanges. The methods of calculation differ, as some use the weighted-volume average and others use a simple average, leading to different daily prices.

¹² Compared to a previous version of this study examining data up to September 2017, the inclusion, in this version, of the bubble boom and burst period going from October 2017 to April 2018, does not show a significant difference in the results of dependence, diversification, and hedging effectiveness analyses.

¹³ The nine series of returns consist of four series in sample I (BTC, S&P500, Gold and Crude Oil) and five series for sample II (BTC, ETH, Gold and Crude Oil). It is important to note that ETH data are only available for sample II. All the descriptive statistics analysis is based on sample II.

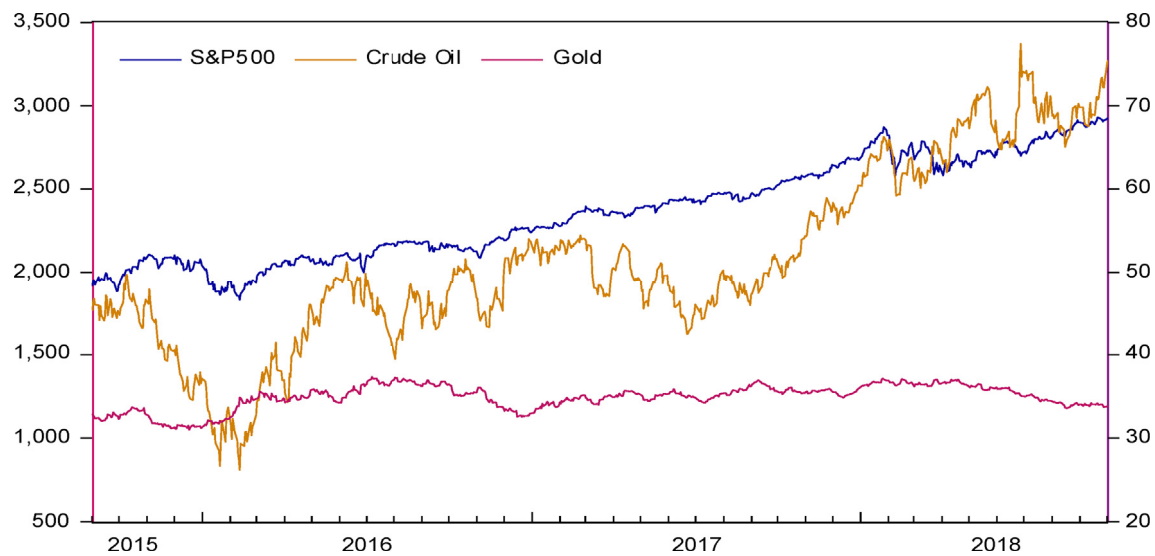


Fig. 3. S&P500, Crude Oil, and Gold price's evolution.

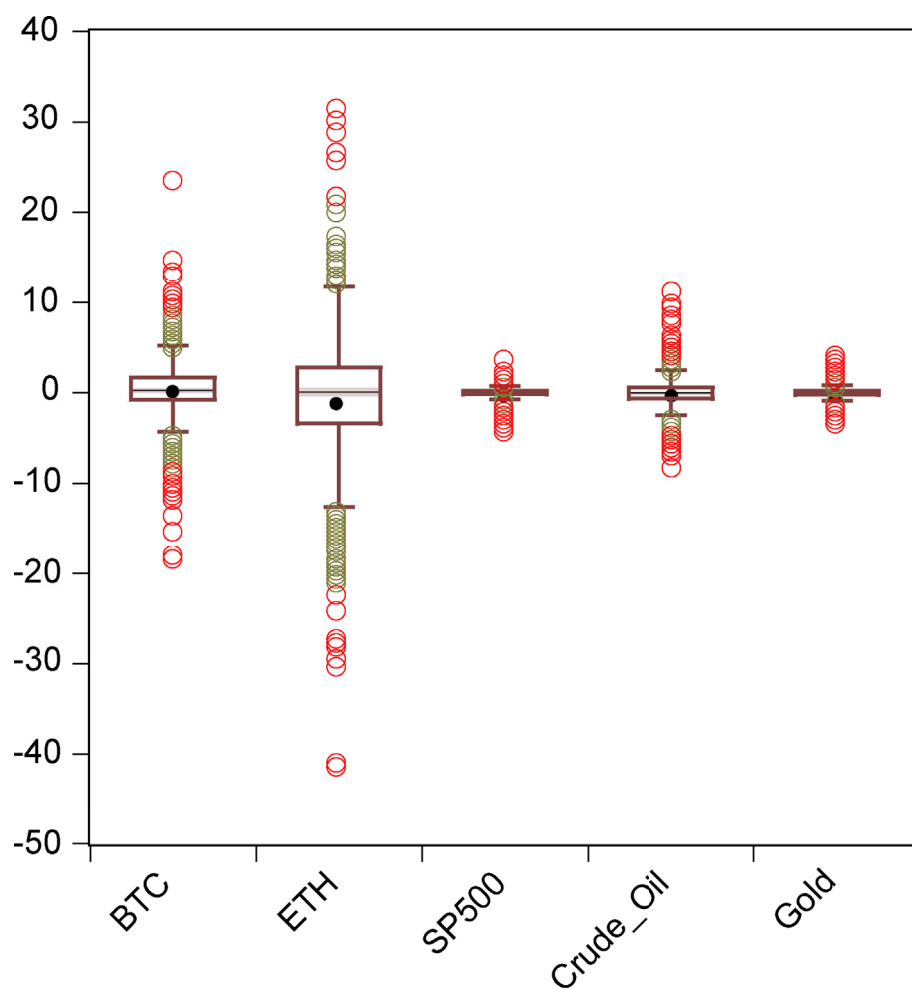


Fig. 4. Box Plot of BTC, ETH, S&P500, Crude Oil, and Gold returns.

Table 2 shows that cryptocurrencies experience the highest increases in prices, followed by crude oil and by the two others conventional financial assets. Using the ratio risk/reward as a criteria of investment, the coefficients of variation, not reported here, show that the digital finan-

cial assets are riskier than conventional financial assets, for the same expected returns.

Additionally, we also find evidence of excess kurtosis for all the considered series of returns, indicating that the distributions of the digital

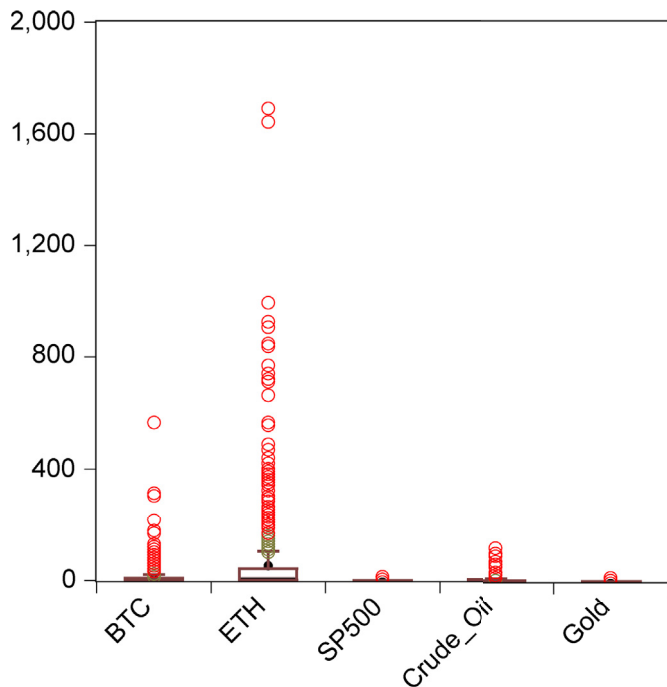


Fig. 5. Box Plot of BTC, ETH, S&P500, Crude Oil, and Gold variances (Squared returns).

and conventional financial assets' returns are fat tailed. Furthermore, the Jarque-Bera test of the normality of the unconditional distribution of each digital and conventional asset strongly rejects unconditional normality in both samples. To test for serial correlation in both the level and return series, we apply the Ljung-Box (LB) statistics for the return and the squared return of the digital and conventional series. The LB statistic on the return series shows that only 4 out of the 9 series show evidence of serial correlation at the conventional level, i.e., 5%. In contrast, the LB statistic on the squared return series reveals a high dependence in the long horizon, suggesting the presence of ARCH effects. This result is confirmed by the ARCH LM statistic, indicating that the returns exhibit conditional heteroskedasticity.

We complete the descriptive analysis by investigating the stationarity and long-range dependence hypotheses. The results of testing for these two properties in the series of returns are reported in Panels A and B of Table 3, respectively. We also report in Panel C the results of testing for long range dependence in the volatility series, proxied by the squared returns. The results of applying the three unit root tests (ADF, PP, and KPSS)¹⁴ on the return series reported in Panel A clearly show that the hypothesis of stationarity of the series of returns cannot be rejected at the conventional level of significance, i.e., 5%. For the long range dependence hypothesis, the results of all the employed tests, i.e., the GSP, ELW, R/S, and V/S,¹⁵ show that the null hypothesis of short memory is only rejected for few cases for the returns series and is highly rejected for all the volatility series. These results are consistent with previous studies showing strong evidence of a long memory in the volatility series but weak evidence in the series of returns (see, for instance, Charfeddine and Benlagha (2016), Guesmi et al. (2018), and Charfeddine and Maouchi (2019)).

In order to visualize and compare the dependence patterns between cryptocurrencies and conventional assets, we consider different pairs

of pure digital, pure conventional and digital-conventional assets. We report in Fig. 6 the scatter-plots, regression lines and the unconditional Pearson correlation coefficients for all the considered pairs. We observe a moderate and positive relationship between Bitcoin and Ethereum with some extreme dependence patterns. For the six digital-conventional financial assets pairs, the scatter plots indicate a low dependence in the center of the distribution, and absence of tail dependence patterns. For the conventional pairs, we observe some extreme dependence patterns. Furthermore, a strong and positive relationship between S&P500 and Gold is highlighted. Overall, the scatter plots reveal significant differences in the dependence patterns among the examined pairs. Consequently, a deeper analysis is needed to get more informative results on the dependence between the different pairs of assets.

We complement the above graphical dependence analysis by reporting in Table 4 the calculated values of five¹⁶ measures of dependence. The results show that none of the 30 calculated coefficients, for the different pairs of digital-conventional financial assets' returns, is statistically significant (see columns 2–4 and 6–8 of Table 4). This result is particularly interesting for financial investors seeking to diversify their portfolios. It suggests that investors can form a low risk portfolio by including one digital and one conventional asset in their portfolio. Regarding the cryptocurrency pair (BTC, ETH), we find that the estimated parameter of dependence is positive and significantly different from zero, no matter the measure of dependence considered. This result offer a primary evidence that the combination of Bitcoin and Ethereum in the same portfolio may reduce the benefits of diversification.

In a nutshell, the preliminary analyses show that while the digital and conventional financial assets share some common stylized facts, they behave differently in many aspects (returns, volatility, dependence patterns). These conclusions reinforce the need to further explore the economic and financial potential of these new assets.

5.2. Marginal distributions analysis and comparison

The results of estimating the $ARFIMA(p, d_m, q)$ - $FIAPARCH(k, d_v, l)$ models for the digital and conventional financial series of returns for the two samples I and II are reported in Table B1 in the supplementary document. The estimation results of the conditional mean equation, reported in Panel A, show that for all the nine series of returns considered in this study, neither the fractional long memory parameter, d_m , nor the AR and MA components are statistically significant. Consequently, only the estimated constant coefficients are reported in Panel A. The above-obtained results are as expected, overall. For instance, as shown in the empirical findings of Panel B of Table B1, there is an absence of long-range dependence in most of the return series. These findings confirm the conclusions of recent studies showing that the cryptocurrencies markets become more efficient with recent data and that the hypothesis of market efficiency (EMH) cannot be rejected for most of cryptocurrencies when using recent data (Brauneis and Mestel (2018); Charfeddine and Maouchi (2019); Tiwari et al. (2018)).

The results of the conditional variance are reported in Panel B of Table B1. The results show that the $FIAPARCH(1, d_v, 1)$ model was selected for all the two cryptocurrency and the $FIGARCH(1, d_v, 1)$ model was adopted for all the conventional financial assets. In particular, the results show that the two cryptocurrencies have the highest fractional long memory parameters, e.g. $d_v = 0.763$ and $d_v = 0.740$ for BTC in samples I and II and $d_v = 0.839$ for ETH, then followed by Crude

¹⁴ The ADF, PP and KPSS refer to the Augmented Dickey-Fuller, Phillips and Perron and Kwiatkowski-Phillips-Schmidt-Shin unit root tests, respectively.

¹⁵ the GSP, ELW, R/S, and V/S refer to the Gaussian semi parametric, exact local Whittle, modified rescaled range, and rescaled variance methods, respectively.

¹⁶ The five measures of dependence include, the unconditional Pearson's rho, Spearman's rho correlation, Kendall's tau, the detrended cross correlation analysis (DCCA), and the constant conditional correlation coefficient of the bivariate GARCH model in Table 4.

Table 2

Descriptive statistics of the returns of the digital and conventional financial assets by sample.

	Sample I: From 18/07/2010 to 01/10/2018				Sample II: From 01/09/2015 to 01/10/2018				
	BTC	S&P500	Gold	Crude Oil	BTC	ETH	S&P500	Gold	Crude Oil
Mean	0.162	0.014	−0.006	0.010	0.130	0.200	0.012	0.000	0.038
Median	0.062	0.000	0.000	0.000	0.121	0.000	0.000	0.000	0.000
Max	18.439	2.012	2.101	4.903	9.833	14.321	1.062	1.822	4.903
Min	−21.347	−2.128	−2.616	−3.931	−7.947	−13.316	−1.662	−1.339	−3.455
Std. dev.	2.503	0.283	0.318	0.666	1.723	2.983	0.239	0.265	0.744
Skew.	−0.345	−0.350	−0.323	0.358	−0.151	0.173	−0.743	0.616	0.762
Kur.	14.964	12.733	12.155	11.080	7.191	6.318	11.195	10.985	10.315
J-B	17,941.12	11,893.77	10,523.04	8220.164	829.142	522.640	3257.58	3065.677	2621.608
P-value J-B	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Q(12)	60.928***	31.717***	18.421	25.152***	6.708	22.547**	20.761*	20.293*	13.013
Q ² (12)	808.89***	880.97***	213.51***	538.18***	185.98***	146.65***	130.65***	71.800***	223.65***
ARCH(12)	37.800***	55.628***	15.788***	34.370***	8.409***	7.533***	9.046***	5.172***	15.111**
Number of obs.	2998	2998	2998	2998	1127	1127	1127	1127	1127

Notes. This table presents the descriptive statistics for the daily return series of Bitcoins, Ethereum, S&P500, Gold and Crude Oil for the two Samples I and II. Std. dev., Skew. and Kurt. refer to the values of standard deviation, the skewness and kurtosis coefficients. J-B is the Jarque-Bera test statistic of normality. For this statistic, the normality hypothesis is rejected if the J-B statistic is greater than $\chi^2(2) = 5.99$. Q(12) and Q²(12) are Ljung-Box tests for 12th - order serial correlation in the returns and squared returns. ARCH (12) is the test for heteroscedasticity for the 12th-order ARCH.

*** Indicates statistical significance at the 1% level.

** Indicates statistical significance at the 5% level.

* Indicates statistical significance at the 10% level.

Table 3

Unit root and long memory tests results.

	Sample I: From 18/07/2010 to 01/10/2018				Sample II: From 01/09/2015 to 01/10/2018				
	BTC	S&P500	Gold	Crude Oil	BTC	ETH	S&P500	Gold	Crude Oil
Panel A: Unit root tests on the returns series									
Return Series	BTC	S&P500	Gold	Crude Oil	BTC	ETH	S&P500	Gold	Crude Oil
ADF	−53.33***	−56.22***	−56.29***	−58.63***	−33.21***	−32.55***	−34.02***	−35.23***	−34.63***
PP	−53.52***	−56.85***	−56.33***	−58.84***	−33.23***	−32.59***	−34.02***	−35.23***	−34.64***
KPSS	0.346	0.080	0.102	0.257	0.169	0.308	0.254	0.178	0.039
Panel B: Long memory tests on the returns series									
GSP	0.121***	−0.123***	−0.013	−0.050	0.074**	0.096***	−0.020	0.033	−0.054
ELW	0.101*	−0.112**	0.007	−0.032	0.026	0.074*	−0.053	0.020	−0.037
R/S	1.657	0.778	1.307	1.456	1.609	1.739	0.797	1.107	1.164
V/S	0.126	0.033	0.096	0.084	0.116	0.173	0.020	0.065	0.037
Panel C: Long memory tests on the volatility series (squared returns)									
GSP	0.263***	0.387***	0.244***	0.308***	0.318***	0.283***	0.313***	0.237***	0.305***
ELW	0.282***	0.387***	0.200***	0.285***	0.287***	0.200***	0.300***	0.141**	0.306***
R/S	2.765***	2.274***	2.981***	3.469***	2.730***	1.225	2.378***	2.765***	2.361***
V/S	0.784***	0.453***	0.641***	0.919***	0.760***	0.092	0.435***	0.736***	0.679***

Notes. This table reports in panel A the results of the standard unit root tests (ADF, PP and KPSS) on the returns series. It also reports the results of the GSP, ELW, R/S AND V/S tests for the returns and volatility series in Panel B and C, respectively.

The ADF, PP, and KPSS results of the unit root tests correspond to the case of unit root with intercept but without trend.

The GSP, ELW, R/S, and V/S refer to the Student-t semi parametric, exact local Whittle, modified rescaled range, and rescaled variance methods, respectively.

*** Indicates statistical significance at the 1% level.

** Indicates statistical significance at the 5% level.

* Indicates statistical significance at the 10% level.

Oil and S&P500 series with values of d_t equal to 0.535, 0.562, and 0.512 and 0.621, respectively in sample I and II. For all these series, the value of the fractional long-range dependence parameter exceed the value of 0.5 in 7 out of the nine considered series, which is evidence for a nonstationary long memory in the volatility series but with mean reverting. The results the long memory component is stationary, as d_t lies in $(0, 1/2)$, only for the Gold series. These results confirm those of Klein et al. (2018). Moreover, we find that the estimated values of the leverage parameter, γ , is significant for the two cryptocurrencies. Interestingly, we find that the estimated γ parameter has a negative sign, an evidence of an inverse leverage effect indicating that these two cryptocurrencies tend to have higher volatility if the price of the previous day increased, a property that is commonly observed in many commodities.

5.3. Dependence patterns between digital and conventional financial markets

The dependence structure between cryptocurrencies and established conventional financial assets, as well as its evolution over time, especially during periods of financial instability and economic turmoil, is a key component to grasp the role of these new digital assets in the economic and financial landscape (see Corbet et al. (2018), Sim and Zhou (2015), Bouri et al. (2017b), Symitsi and Chalvatzis (2018), and Wei et al. (2018)). To investigate this issue, we use the static and time varying copula techniques to estimate the dependence dynamics between the different digital and conventional financial assets, as well as the digital assets with each other. Overall, the empirical results show that the Student-t copula outperforms all the other six consid-

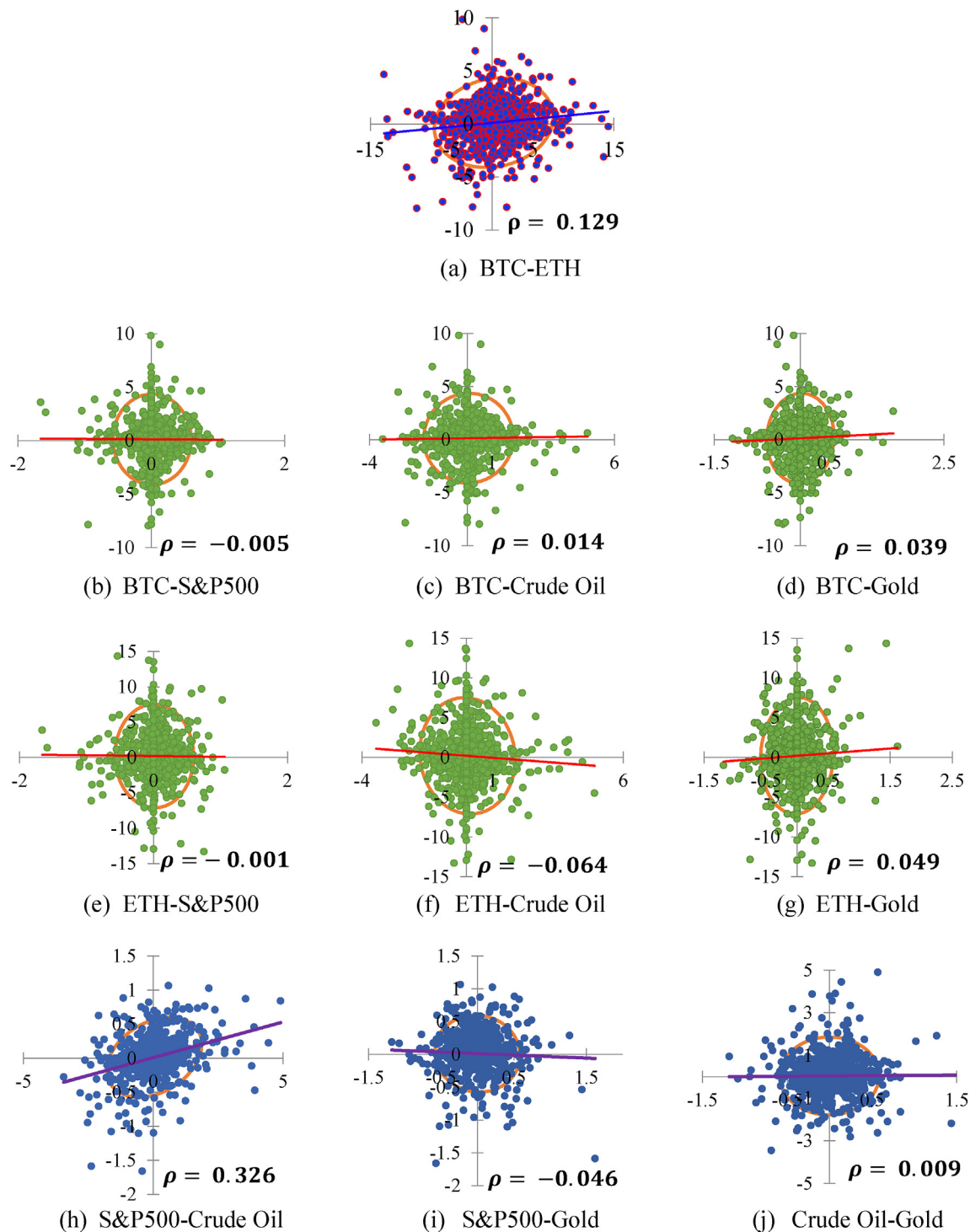


Fig. 6. Scatter Diagram for the ten returns pairs.

ered copulas. The results show also, with clear evidence, that the time varying dependence parameter is weak over all the period of analysis and for all the pairs examined, except for the BTC-ETH pair.¹⁷ To facilitate the analysis of the cross-correlation pattern between the different considered financial assets pairs, we plot them in Fig. 7 through Fig. 13. We also report the different breaks dates and the mean of each time-varying dependence parameter regime, as identified using the

Bai and Perron (2003) tests for structural breaks. The results show evidence of structural breaks existence in the returns of all pairs without any exception. All the graphics clearly show that the parameter of dependence is changing over time. In what follows, we analyze this time varying dependence by first examining the digital-S&P500 pair, followed by the digital-Gold pair, the digital-Crude Oil pair, and finally the BTC-ETH pair.

The time-varying patterns of the two BTC-S&P500 and ETH-S&P500 pairs are plotted in Figs. 7 and 8, respectively. We observe that the dependence parameter is awfully volatile and alternating between pos-

¹⁷ The discussion of the different estimated static and time varying copulas is reported in the [supplementary document](#) of this paper.

Table 4
Estimation results of different static measures of dependence between two time series.

	BTC			BTC ETH	ETH		
	S&P500	Gold	Crude Oil		S&P500	Gold	Crude Oil
Pearson's Rho	0.010	0.017	−0.008	0.129***	−0.001	−0.049	−0.064
Spearman's Rho	0.008	0.036	0.003	0.104***	−0.008	0.013	−0.021
Kendall's tau	0.006	0.026	0.007	0.070***	−0.007	0.009	−0.014
DCCA	0.022	0.021	0.009	0.144***	−0.003	0.021	−0.055
CCC - GARCH	0.003	−0.029	−0.014	0.138***	0.028	0.034	−0.027

Notes. This table reports the results of the estimation of the different static measures of dependence for all the seven returns pairs considered in our study.

*** Indicates statistical significance at the 1% level.

** Indicates statistical significance at the 5% level.

* Indicates statistical significance at the 10% level.

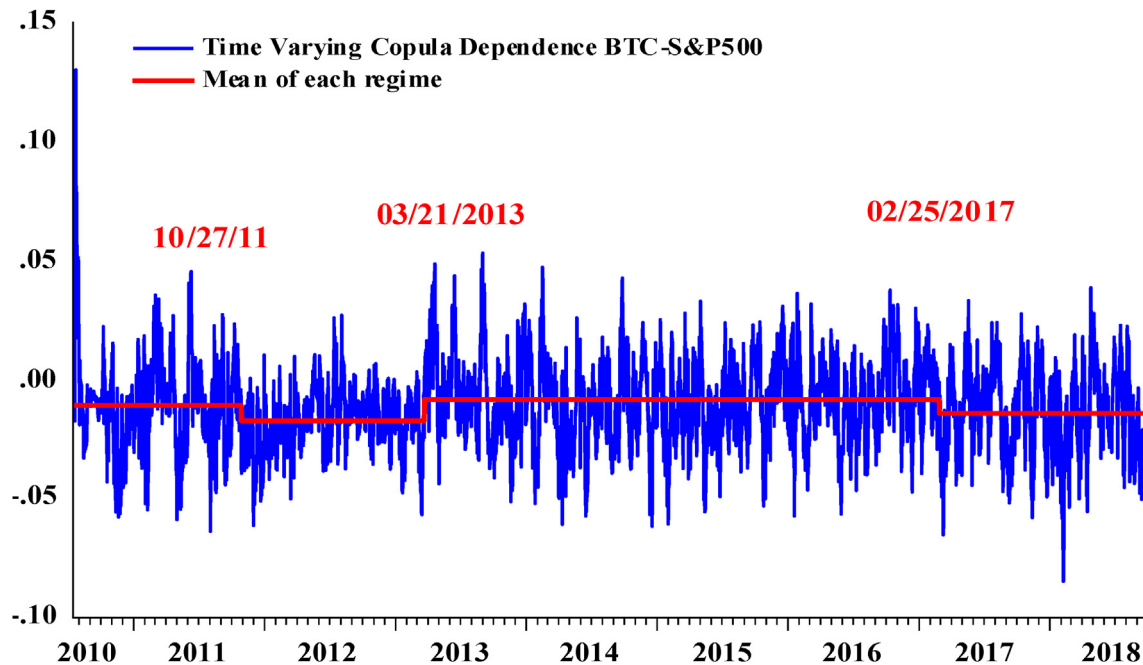


Fig. 7. Time varying dependence parameter from the Student-*t* copula for the pair BTC-S&P500.

itive and negative values. However, when looking into the behavior of the relationships, we observe that the average dependence between BTC and S&P500 is negative (a mean of −0.001) and positive between ETH and S&P500 (a mean of 0.016), for the whole period.

More interestingly, the figures show different regimes of dependence levels between the cryptocurrencies and S&P500. These regimes are quite similar when comparing the BTC-S&P500 and ETH-S&P500 pairs. Precisely, four regimes are highlighted for both pairs. After a period of relatively high volatility in the dependence between the returns of cryptocurrencies and S&P500, their co-movement becomes steady for the period 2011–2013. After that, a third cluster appears indicating higher level of dependence between the returns of the considered pairs. Finally, starting from 2015, the co-movement between the studied assets becomes extremely volatile but significantly lower than the foregoing periods. These different regimes in the dependence between cryptocurrencies and S&P500 are closely related to various economic and financial events detailed in Figs. 14 and 15. We first notice a change around October 2011, where the dependence between BTC and S&P500 slightly decreases, which coincides with the Eurozone debt crisis discussion between Greece and the tripartite committee formed by the European Commission, the European Central Bank and the International Monetary. The dependence rises again in March 2013 in parallel with the Cyprus debt crisis. As for the relationship between ETH and S&P500, the first change occurs in May 2016 where

the relationship decreases almost to zero. The main event registered on May 2016, is the crowdfunding campaign using Ethereum of the Decentralized autonomous organization (DAO), which set the record for the largest crowdfunding campaign in history with 120 Millions USD worth of Ether raised. Toward the end of the investigated period, we see a common decrease of dependence between the two cryptocurrencies and equities proxied by the S&P500 index. These observations could be linked to what happened in the respective BTC and ETH markets, such as the rejection of the first BTC Exchange traded funds by the U.S. Securities Exchange Commission (SEC) or the major Chinese cryptocurrencies exchanges halting withdrawals, following the Chinese central bank's warning regarding the need to enforce rules on money-laundering and foreign exchange.

Regarding the two pairs BTC-Gold and ETH-Gold, the relationship seems to be more complicated than that of the two cryptocurrencies-S&P500 cases. We observe that the average of dependence is positive for the two pairs BTC-Gold (a mean of 0.028), and ETH-Gold (a mean of 0.034). We also observe from Figs. 9 and 10, as well as Table 5, that the dependence between Ethereum and Gold is more volatile, as it varies between a minimum of −0.102 and a maximum of 0.134, while Bitcoin's dependence parameter with Gold moves between 0.003 and 0.051. Similar to its relationship with S&P500, Bitcoin's correlation with Gold, decreases starting from March 2017. Ethereum on the other side sees and increase of its dependence with Gold, contrary to

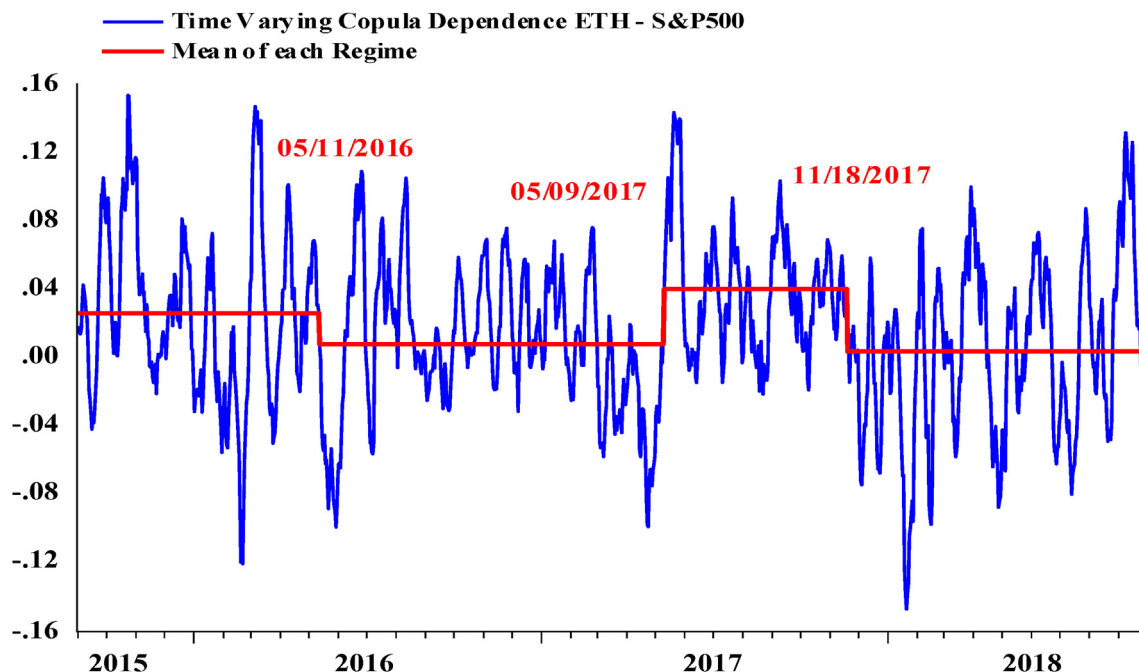


Fig. 8. Time varying dependence parameter from the Student-*t* copula for the pair ETH-S&P500.

its relationship with S&P500, for the same period. This opposite behavior is also observed for the whole sample II, in which the respective relationships between each cryptocurrency and Gold, moves on average in opposite directions. Moreover, even if we do not see any stable correlation between the two cryptocurrencies and Gold, we however observe that the changes in the dependence regimes happened around some important event in the CC or Gold Market. The break identified late April 2013, showing a decrease of dependence between BTC and Gold, occurs during the Cyprus debt crisis, where officials discussed tapping into their excess reserves of gold to contribute to the country's bailout. The increase of the BTC-Gold dependence in July 2014, happens in parallel with the forum organized by the World Gold Council organized to debate reforms of the London Gold Fix. Regarding Ethereum's dependence with Gold, the Annual Ethereum conference during which Microsoft and Thompson Reuters unveiled projects based on Ethereum, or the Cyber attack that lead the Ethereum network to run slowly seems to have had significant impacts on the relationship.

The dynamic dependence between BTC-Crude Oil and ETH-Crude Oil pairs are reported in Figs. 11 and 12, respectively. The results of the descriptive statistics of these two time series, reported in Table 5, show that they are characterized approximately by equal negative means and medians indicating absence of outliers in their dependencies dynamics. In particular, we find that they have maximums of dependence of 0.104 and 0.058, and minimums of dependence of -0.097 and -0.138 , respectively. Moreover, the results of the Bai and Perron tests show these two dependence patterns were subject to three breaks and one break, respectively. For the BTC - Crude Oil dependence, we find that the break date of October 2011 coincide with Eurozone debt deal discussion. The February 2015, break does not correspond to a particular event, beside the fact that it corresponds to the period (January–February 2015) in which oil prices registered their first recovery from the plunge that affected oil markets, leading the price of the barrel to decrease from approximately 106 dollars on July 2014 to approximately 45 dollars on January 2015. Regarding the ETH - Crude Oil dependence, we find a single break date, in October 2016, which corresponds to the first fork of Ethereum that lead to the creation of Ethereum classic (ETC) in parallel to Ethereum. This break also coincides with the announcement by J.P. Morgan to build a transaction

platform that will be based on ETH. On the oil market side, this break comes at the time of Russia's announcement to back OPEC decision to freeze its production. Following the announcement, oil prices witnessed a 2.5% increase.

Turning to the two cryptocurrencies, we note that the dependence structure is different from the previous ones, with two important observations. The results show that the parameter of dependence between the two cryptocurrencies is overall high compared to other assets cases. Moreover, the dependence between the two digital assets is increasing over time but does not exceed the 0.2 level on average. Regarding, the increasing dependence, it should be noted that our sample goes back to the creation of Ethereum, therefore within the considered period, this cryptocurrency gained in term of acceptance and maturity, which could be behind the increase of its dependence with Bitcoin. Indeed, based on the dependence pattern (Fig. 13), we observe that the average dependence between Bitcoin and Ethereum was close to 0 from the creation of Ethereum to mid-2016. Starting from mid-2016 the dependence started to increase to reach 0.1 in mid-2017 and 0.2 in mid-2018. We also observe that the last period (starting from March 2018) shows a co-movement with a lower volatility than the period prior to March 2018. One would assume that the correlation between Bitcoin and Ethereum, would be higher compared to the results obtained. However, considering the nature of the two cryptocurrencies, these results can make sense. As highlighted in Sec.3, despite their similarities (mined blockchain-based cryptocurrencies), Bitcoin and Ethereum are quite different. Indeed, Bitcoin is a pure cryptocurrency, while Ethereum can be viewed as a single shared computer, and "Ether" represent the currency that can be used to power the smart contracts on the platform. Therefore the usage of Ethereum is far richer than Bitcoin. For instance the Initial Coins Offerings are a proof of this usage, as almost all the ICOs are based on Ethereum. Moreover, these results are in line with the recent research on cryptocurrencies (see Baumöhl (2018); Ciaian et al. (2018)).

Overall, the above analysis shows that the dependence between digital and conventional assets is very weak, and sensitive to external shocks and events. Due to the weakness of their relationship with established conventional assets, cryptocurrencies appear to have a role in helping investors generate profit by adding a digital asset to a portfolio of conventional assets. To better characterize these investment opportu-

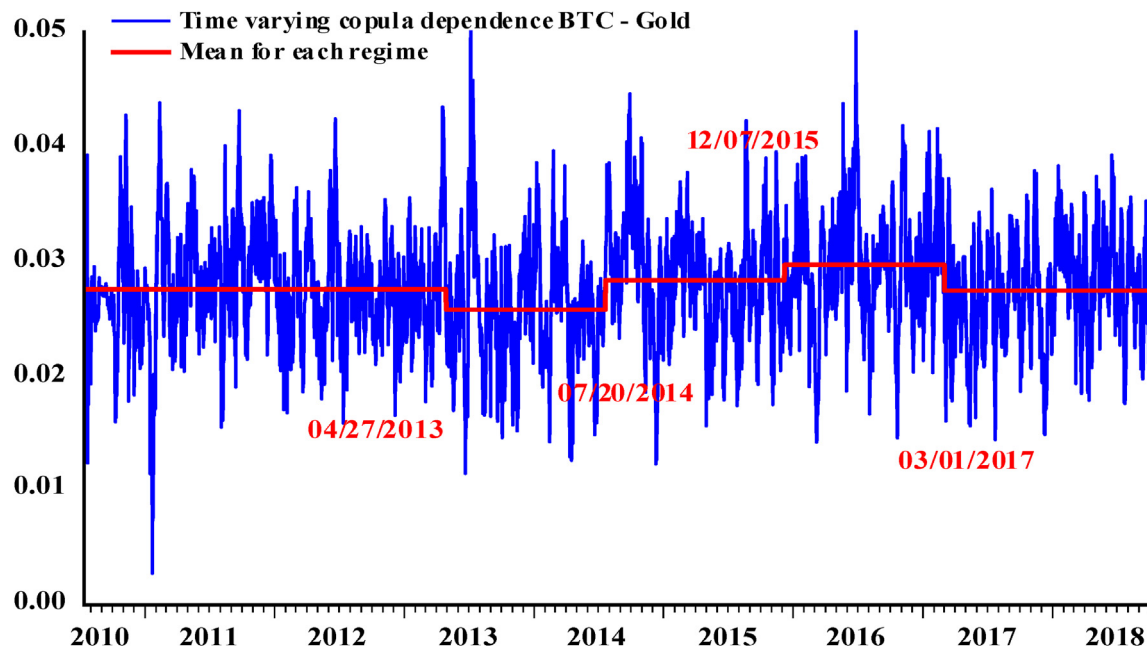


Fig. 9. Time varying dependence parameter from the Student- t copula for the pair BTC-Gold.

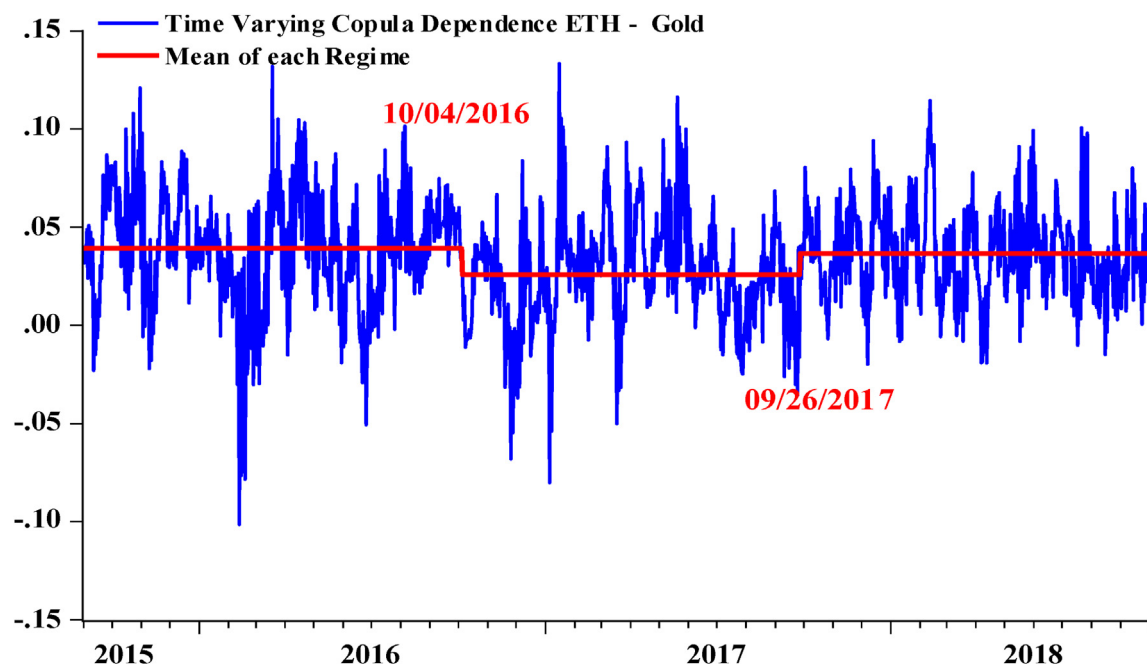


Fig. 10. Time varying dependence parameter from the Student- t copula for the pair ETH-Gold.

nities, we construct and carry out analyses of different portfolios including cryptocurrencies.

6. Implications for portfolio management and hedging performances

In light of the empirical findings, discussed in Sec. 5.3, one may expect generating substantial gains from portfolios combining digital and financial assets, or exclusively composed of digital financial assets. To quantify these gains, we calculate in this section the optimal weights, hedging ratios, and measure the hedging effectiveness of seven portfolios.

We begin our analysis by considering the case of an investor who construct a hedged portfolio containing one digital and one conventional financial asset. The investor wants to protect his exposure to the conventional financial assets' (S&P 500, Gold, and Crude Oil) price movements by investing in a CC (Bitcoin and Ethereum). The objective of the investor is to minimize the risk of his crypto-financial assets portfolio while keeping its expected returns the same. We also investigate the opportunities offered by exclusively investing in digital financial assets (BTC and ETH).

Following Kroner and Ng (1998), the optimal weights of a crypto-financial asset portfolio at time t , subject to no-short selling, denoted

Table 5
Statistics of the estimated time varying parameter of dependence.

	BTC			BTC	ETH		
	S&P500	Gold	Crude Oil	ETH	S&P500	Gold	Crude Oil
Mean (M)	−0.019	0.028	−0.003	0.101	0.016	0.034	−0.021
Median	−0.019	0.028	−0.003	0.099	0.016	0.033	−0.020
Max	0.130	0.051	0.104	0.537	0.153	0.134	0.058
Min	−0.085	0.003	−0.097	−0.274	−0.148	−0.102	−0.138
Std.Dev.	0.019	0.005	0.025	0.126	0.048	0.029	0.027
Skewness	0.261	0.095	−0.009	0.269	−0.039	−0.145	−0.324
Kurtosis	4.473	3.616	3.507	3.546	3.288	4.030	3.703
J-Bera	312.22	51.839	32.142	26.733	4.181	53.762	42.924
p – value	0.000	0.000	0.000	0.000	0.124	0.000	0.000
t-stat (M = 0)	−34.5170	286.3647	−6.5822	26.975	11.718	39.127	−25.452
W	237.544	246.335	233.471	65.840	82.254	86.813	87.346
A-D	1109.203	1145.564	1092.653	328.569	388.821	407.735	409.352

Note. This table displays the descriptive statistics of the time varying estimated values of the parameter of dependence using the Student-*t* copula. W and A-D refer to the Watson and Anderson-Darling tests, respectively.

*** Indicates statistical significance at the 1% level.

** Indicates statistical significance at the 5% level.

* Indicates statistical significance at the 10% level.

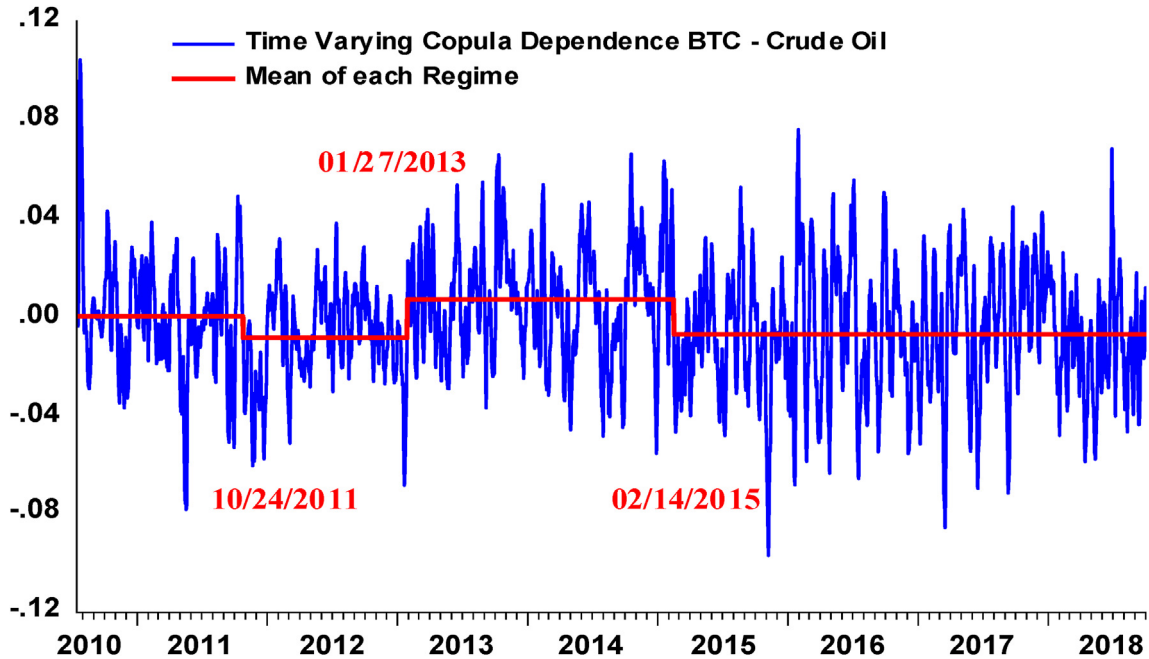


Fig. 11. Time varying dependence parameter from the Student-*t* copula for the pair BTC-Crude Oil.

W_{ijt} is defined by the following:

$$W_{ijt} = \frac{h_{j,t} - h_{ij,t}}{h_{i,t} - 2h_{ij,t} + h_{j,t}} \quad (9)$$

where $h_{i,t}$, $h_{j,t}$, $h_{ij,t}$ refer respectively to the conditional variance of the digital financial asset d_i returns, the conditional variance of the conventional financial asset c_j returns and the conditional covariance between the digital asset d_i and the conventional asset c_j returns at time t with i = Bitcoin or Ethereum and j = S&P 500, Gold, or Crude Oil.

Assuming a mean-variance utility function, the optimal portfolio holdings of digital asset i is given by the following:

$$W_{ijt}^* = \begin{cases} 0, & \text{if } W_{ijt} < 0. \\ W_{ijt}, & \text{if } 0 \leq W_{ijt} \leq 1. \\ 1, & \text{if } W_{ijt} > 1. \end{cases} \quad (10)$$

and the optimal portfolio holding of the conventional financial asset j is $1 - W_{ijt}^*$.

To optimally hedge the risk of his investment in conventional assets, the investor should take an appropriate position on the cryptocurrencies market to minimize the risk of the hedged portfolio. A long position (buying) of one dollar on the conventional financial asset j must be hedged by a short position (selling) of β_{ijt} dollars on the digital asset i . According to Kroner and Sultan (1993), the optimal hedge ratio is given by the ratio of the conditional covariance between the returns of the digital and the conventional assets, $h_{ij,t}$, to the variance of the digital asset returns, $h_{i,t}$, at time t , as follows:

$$\beta_{ijt}^* = \frac{h_{ij,t}}{h_{i,t}} \quad (11)$$

Moreover, in order to better assess the hedging capabilities of CC, we calculate their hedging effectiveness. To do so, we construct two types of portfolios: (1) unhedged portfolios, exclusively formed of conventional financial assets, and (2) hedged portfolios that are formed by

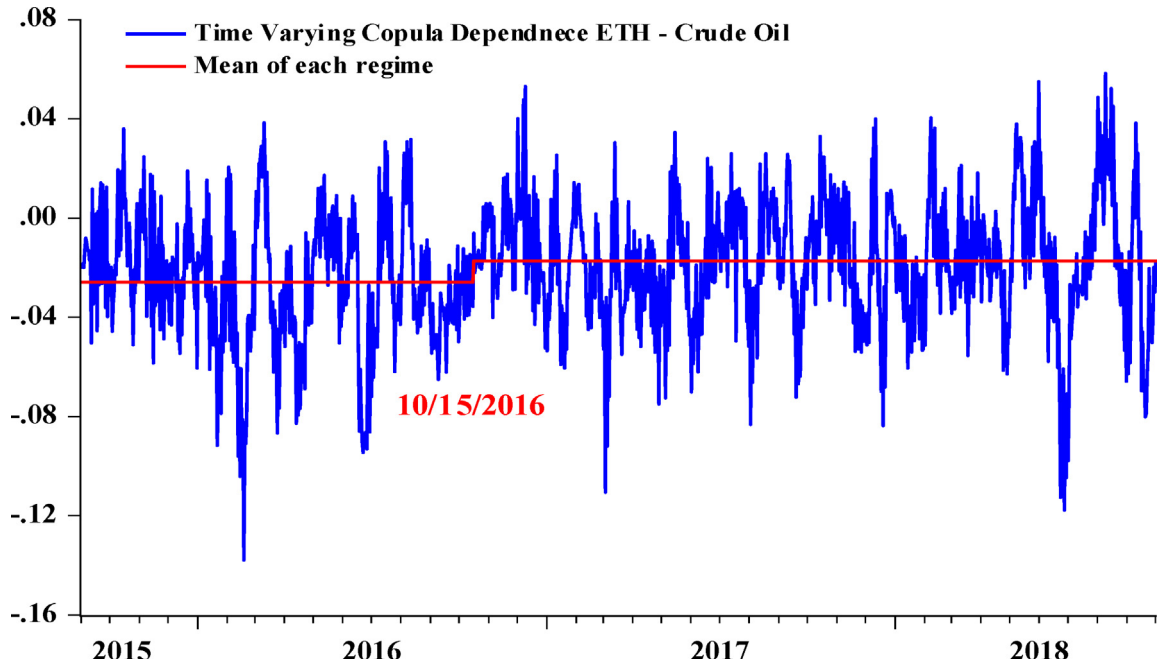


Fig. 12. Time varying dependence parameter from the Student-*t* copula for the pair ETH-Crude Oil.

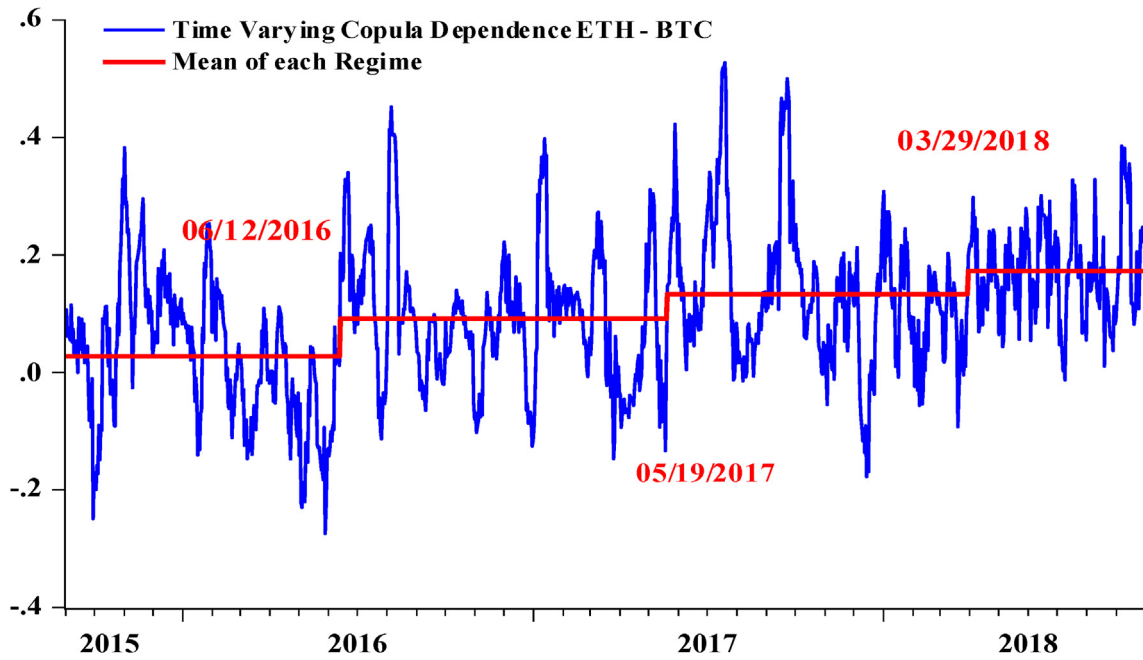


Fig. 13. Time varying dependence parameter from the Student-*t* copula for the pair BTC-ETH.

investing W_t^* in digital assets and $1 - W_t^*$ in conventional assets. We calculate the weights, W_t^* used in our analysis from Eq. (10). Finally, we compute the hedging effectiveness (HE) index as:

$$HE = \frac{Var_{unhedged} - Var_{hedged}}{Var_{unhedged}} \quad (12)$$

where Var_{hedged} and $Var_{unhedged}$ refer to the variance of the hedged portfolio and the variance of the unhedged portfolio, respectively. A higher HE index indicates a higher hedging effectiveness and a larger risk reduction.

The calculation results of the optimal weights average values W_{ijt} and optimal hedge ratios β_{ijt}^* using Gaussian and Student-*t* copulas are

reported in Table 6. For robustness checking, we report also the results based on the BEKK-GARCH, DCC-GARCH and ADCC-GARCH models in Table 7.

We observe from the results that except for the pairs (BTC-Crude Oil), (BTC-ETH) and (ETH-Crude Oil) for which the optimal weights of cryptocurrencies in the investor portfolio range from 11.62% to 24.25%, the optimal weights of cryptocurrencies for the remaining pairs are always below 7%, regardless of the method used. Overall, we find that to minimize the risk while keeping the same expected returns of the digital-conventional financial asset portfolio, the investor should hold more conventional financial assets than digital assets. This empirical finding can be explained by the low dependence between returns of the

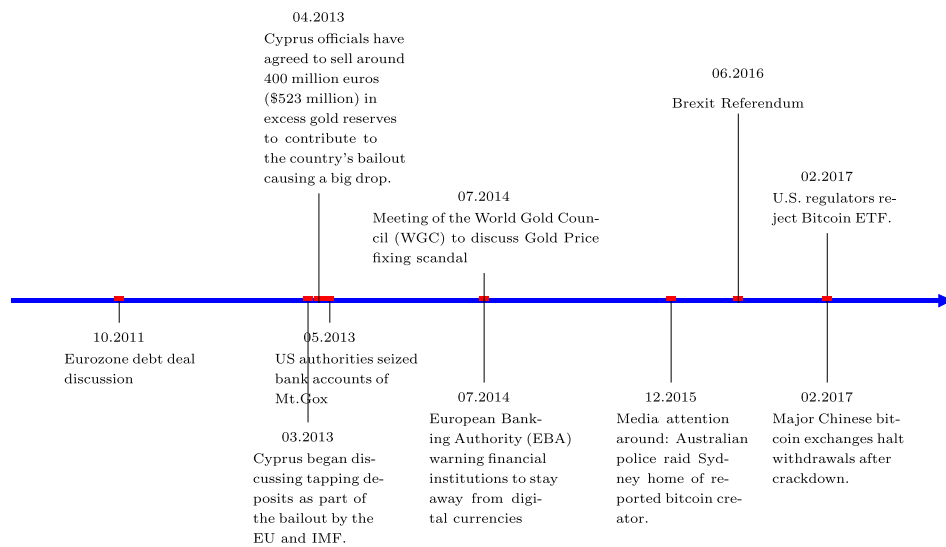


Fig. 14. Events surrounding break dates for BTC, S&P500, Gold, and Crude Oil.

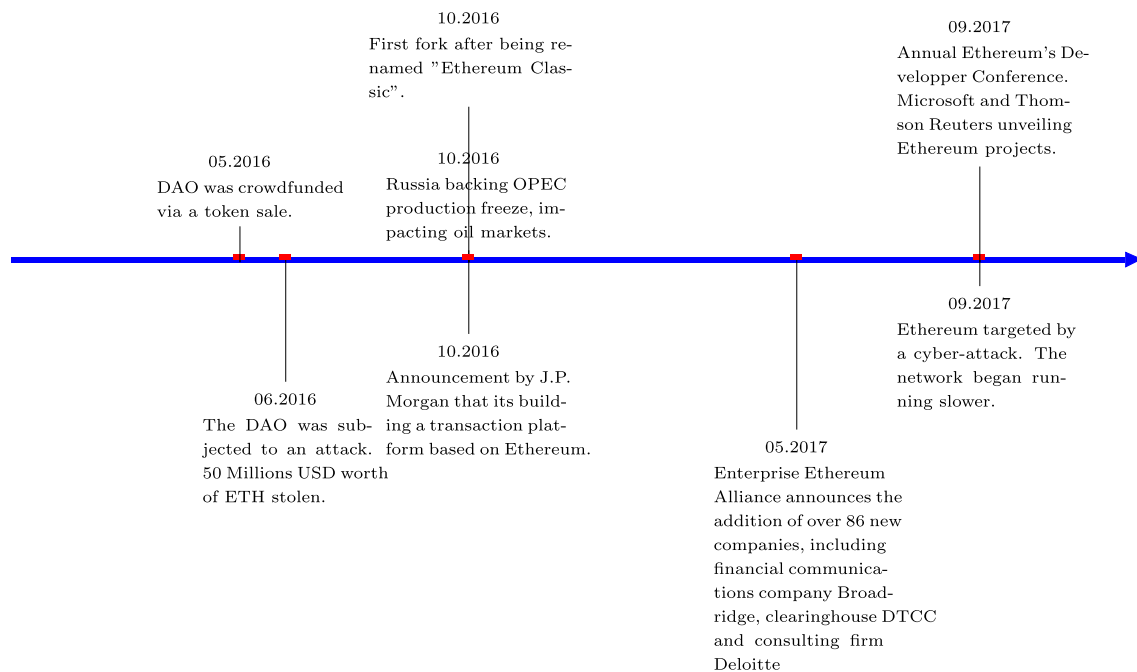


Fig. 15. Events surrounding break dates for ETH, S&P500, Gold, and Crude Oil.

digital and conventional assets, as well as by the high volatility that characterizes digital asset prices. These results imply that adding a very small portion of digital assets to a diversified portfolio of conventional financial assets will substantially reduce its overall risk for a given level of expected return.

As for the optimal hedge ratio β_{jt} , all the models report very low values for most pairs, less than 2.5%, and negative values for the BTC-S&P 500, BTC-ETH and ETH-Crude Oil pairs. These results have two important implications. First, the low values of the hedge ratios suggest that the investment risk of conventional financial assets can be hedged by taking short positions in the cryptocurrencies market. For instance, using the DCC GARCH model, a one dollar long position on S&P 500

should be hedged by going short 0.003 USD on Ethereum. Second, the negative hedge ratio of the BTC-S&P 500 and BTC-ETH pairs implies that to hedge a one dollar invested in S&P 500 or Ethereum, an investor should buy between 0.001 and 0.004, and between 0.012 and 0.029 of BTC, respectively.

Regarding the hedging effectiveness, we observe that hedging strategies involving digital and conventional financial assets moderately reduce the portfolio's risk (variance) in the case of BTC-Crude Oil, with a variance reduction ranging from 28.30% in the case of the time varying Gaussian copula to 32.17% when using the time varying Student copula model. The portfolio's risk can also be reduced in the case of BTC-ETH, with a hedging effectiveness index ranging from

Table 6
 W^d , $\beta^{d,c}$ and HE using copula dependence models.

	BTC			BTC ETH	ETH		
	S&P500	Gold	Crude Oil		S&P500	Gold	Crude Oil
Gaussian							
W^d	0.048	0.056	0.184	0.240	0.023	0.053	0.118
$\beta^{d,c}$	−0.002	0.006	−0.002	−0.012	0.002	0.002	−0.018
HE	0.095	0.093	0.283	0.231	0.028	0.008	0.099
Student							
W^d	0.049	0.057	0.174	0.251	0.025	0.036	0.125
$\beta^{d,c}$	−0.003	0.006	0.013	−0.028	−0.048	0.017	−0.002
HE	0.092	0.094	0.322	0.246	0.019	0.031	0.101

Notes: The table reports average optimal weight of digital asset and hedge ratios for a digital-conventional financial asset portfolio using Copula approach.

Table 7
 W^d , $\beta^{d,c}$ and HE using bivariate GARCH models.

	BTC			BTC ETH	ETH		
	S&P500	Gold	Crude Oil		S&P500	Gold	Crude Oil
BEKK							
W^d	0.057	0.058	0.204	0.234	0.023	0.044	0.128
$\beta^{d,c}$	−0.005	0.002	0.002	−0.024	0.001	0.004	−0.013
HE	0.048	0.057	0.291	0.219	0.027	0.007	0.074
DCC							
W^d	0.053	0.068	0.213	0.214	0.014	0.018	0.125
$\beta^{d,c}$	−0.001	0.003	0.001	−0.019	0.003	0.004	−0.012
HE	0.042	0.074	0.287	0.206	0.013	0.008	−0.082
ADCC							
W^d	0.052	0.062	0.202	0.243	0.032	0.023	0.116
$\beta^{d,c}$	−0.004	0.003	0.004	−0.030	0.003	0.005	−0.009
HE	0.0682	0.065	0.292	0.198	0.024	0.008	0.093

Notes: The table reports average optimal weight of cryptocurrencies and hedge ratios for a digital-conventional financial asset portfolio using conditional variances and covariance estimated using multivariate GARCH approach.

19.82% (ADCC-GARCH) to 24.53% for the time varying student-*t* copula. For the rest of the pairs, cryptocurrencies appear to be a poor hedging instrument, as the HE index is always below 7%, except for the cases of ETH-Crude Oil, where the HE index ranges between 7.38% and 10.12%, and BTC-Gold, where the HE index ranges between 5.73% and 9.40%. These empirical findings confirm some previous studies on the hedging capabilities of cryptocurrencies (see for instance, [Guesmi et al. \(2018\)](#), and [Corbet et al. \(2018\)](#)). Overall, the results show that digital assets are, in most cases, better diversifier instruments and can, to some extent, be used to hedge investments in Crude Oil, and in digital assets.

7. Conclusion

This paper investigates the dynamic relationship between cryptocurrencies and a conventional financial assets. In particular, we investigate the economic and financial benefits of two of the leading traded digital assets, Bitcoin and Ethereum, for financial investors. To do so, we assess the capabilities of cryptocurrencies to generate benefits from portfolio diversification as well as hedging strategies. Three important conclusions can be drawn from this study. First, using time varying copula models, we find evidence of time varying dependence between all the different pairs considered. This evidence is confirmed by the three bivariate GARCH models used for checking the robustness of the results. Moreover, the results of estimating static and time-varying tail copulas (Frank's, Joe's and Gumbel's copulas) indicate the absence of any asymmetric tail dependence between the returns of digital and conventional financial assets. Second, we show that for the case of portfolio with mixed assets, the level of dependence is very weak for all the seven examined pairs without exception, a result which suggests that Bitcoin

and Ethereum can offer new opportunities for portfolio diversification. In addition, our results show that, in general, optimal diversification can be reached by including only a small weight of digital assets to a portfolio of conventional financial asset. This result departs from previous studies in the empirical literature that have investigated diversification considering only conventional financial assets. On the other hand, we find that in the case of a portfolio formed by digital financial assets only, the level of dependence is high compared to the case of portfolios with mixed assets. Third, our empirical results show that digital assets hedging capabilities are very weak, with the maximum values of the hedging effectiveness being 32.17% and 29.20%, obtained for the BCT-Crude Oil pair when using the Student-*t* copula approach and the ADCC bivariate GARCH model, respectively.

In addition to all the previous results and conclusions, we also find evidence that the relationship between cryptocurrencies and conventional assets is sensitive to external shocks. Using [Bai and Perron \(2003\)](#) tests for structural changes, we find several structural changes in the time varying parameter of dependence. The identified break dates correspond to several important financial and international events, including the Euro-zone Debt Deal discussions of 2011, the financial crisis that affected Cyprus and the different announcements made by its government and the EU in 2013, or the Brexit referendum in 2016. From the different dates, we also observe that many break points correspond to events relative to the cryptocurrencies markets, such as the Ethereum fork, the DAO's hacking, rumors surrounding key Cryptocurrency sphere figures or announcements by corporations, such as Microsoft or J.P. Morgan, to use the Blockchain technology. We also observe that numerous changes in the relationship between the two categories of assets were identified within few days of important decisions

or announcements by the U.S. Securities and Exchange Commission or the European Banking Authority.

The results of this study have several important implications for investors, market analysts and participants. In addition to helping shed more light on the nature of cryptocurrencies, this study provides a better representation of the dependence between digital and conventional financial assets, which is a key insight for investors and portfolio managers seeking to minimize the risk of their portfolio without compromising the returns.

Practically, we show in this paper that some successful diversification strategies might be identified, and investors will benefit from the diversification capabilities of the cryptocurrencies if the optimal weight combination of digital-conventional assets is used. Indeed, the low values of the diversification weights indicate that only small amounts of CC are necessary to reach an optimal diversification level. This is explained mainly by the very low dependence between digital financial assets and conventional assets and the high levels of volatility of CC. The only exceptions to the low levels of diversification weights are found in the two cases of Crude Oil and the BTC-ETH pair. However, the highest weights found in this study are significantly lower than studies that have only considered conventional financial assets for diversification. This latter result implies that the two digital and the three conventional financial assets have different behaviors. Regarding the hedging capability of CC, we found that BTC and ETH are poor hedging instruments, as their hedging effectiveness index is low compared to the previous studies in the empirical literature only considering conventional financial assets.

A direct implication of this result for investors and market participants is the need to follow the legal evolution of CC markets. Indeed, as mentioned above, the price evolution of cryptocurrencies is, to a large extent, fueled by the legal uncertainties surrounding these markets. A clear legal framework will lead to different behaviors of cryptocurrencies, which would, therefore, impact their diversification and hedging capabilities and their relationship with the existing asset classes.

Despite the several results of this study that contribute to a better understanding of cryptocurrencies and their financial potential, some shortcomings related to the nature of the digital assets are worth mentioning. First, to explore the potential hedging of cryptocurrencies, we used the spot prices. The nascent future markets of cryptocurrencies is a new tool at the disposition of investors not only to hedge cryptocurrencies themselves, but that can be used to hedge other financial assets. However, these markets are still not mature, as the first introduction of Bitcoin futures in the CBOE exchange in Chicago was recent. The democratization of these contracts and their expansion to other cryptocurrencies will change the landscape of financial hedging and diversification using of cryptocurrencies. Second, an investor seeking investment in CC, will probably consider more than Bitcoin and Ethereum. Therefore, there is a clear need to include more digital assets in the analysis. This need is more imperative, considering the pace at which new cryptocurrencies are being introduced and the growing reliability of the existing ones.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.econmod.2019.05.016>.

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